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ABSTRACT

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Naegleria fowleri, a highly pathogenic thermophilic amoeba, causes Primary Amoebic Meningoencephalitis (PAM), a rapidly progressing central nervous system infection with a mortality rate exceeding 97%. The severity of the disease, combined with the limited efficacy and significant toxicity of current treatments, underscores the urgent need for novel therapeutic strategies. In this study, an integrated in silico drug repurposing approach was employed to screen 2,064 compounds against glucokinase (NfGlck), a key enzyme essential for glycolysis and survival of *N. fowleri*. The three-dimensional structure of NfGlck (PDB ID: 6DA0) was retrieved from the PDB and subjected to physicochemical characterization using ExPASy ProtParam, along with functional annotation via InterPro. Phylogenetic analysis using MEGA V12.1 confirmed evolutionary conservation of the target protein. High-throughput virtual screening was conducted using PyRx V0.8 (AutoDock Vina), with binding affinities evaluated against a threshold of –14.9 kcal/mol. Docking reliability was validated through redocking of the native ligand (ANP), ensuring conformational accuracy. Residue-level interaction analysis revealed conserved residues, including GLY56, GLY206, and GLY208, across native and docked complexes. Protein–ligand stability and flexibility were assessed using CABS-flex 2.0, with RMSF profiles indicating stable behavior beyond 500 ns compared to the native system. Further, ADMET profiling evaluated pharmacokinetic properties, while Density Functional Theory (DFT) calculations using the B3LYP functional assessed electronic reactivity through HOMO–LUMO analysis. Overall, seven potential lead compounds were identified, demonstrating strong binding affinity and favourable stability, providing a solid foundation for future experimental validation.

Keywords: *Naegleria fowleri, Primary Amoebic Meningoencephalitis, Glucokinase, In silico drug repurposing, Molecular docking, ADMET profiling, Density Functional Theory*

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ABBREVIATIONS

Abbreviation	Full Form
ADMET	Absorption, Distribution, Metabolism, Excretion, and Toxicity
ANP	Adenosine Nucleotide Analog
ATP	Adenosine Triphosphate
BBB	Blood-Brain Barrier
BGC	β-D-glucose
BLAST	Basic Local Alignment Search Tool
BLASTp	Basic Local Alignment Search Tool for proteins
CASB-flex	Coarse-grained modeling tool for protein dynamics
CDC	Centre for Disease Control
CDD	Conserved Domain Database
CID	Compound Identifier
CNS	Central Nervous System
DFT	Density Functional Theory
DMPNN	Directed Message Passing Neural Network
EC	Enzyme Commission
FMO	Frontier Molecular Orbital
GI	Gastrointestinal
GRAVY	Grand Average of Hydropathicity
HBA	Hydrogen Bond Acceptor
HBD	Hydrogen Bond Donor
HIA	Human Intestinal Absorption
HOMO	Highest Occupied Molecular Orbital
HTVS	High-Throughput Virtual Screening
IV	Intravenous
JTT	Jones-Taylor-Thornton
LUMO	Lowest Unoccupied Molecular Orbital
MEGA	Molecular Evolutionary Genetics Analysis
ML	Maximum Likelihood
NBD	Nucleotide-Binding Domain
NCBI	National Centre for Biotechnology Information
NIM	National Institute of Medicine
NfGlck	Glucokinase of *Naegleria fowleri*
nr	Non-redundant
PAM	Primary Amoebic Meningoencephalitis
PDB	Protein Data Bank
PDBQT	Protein Data Bank, Partial Charge (Q), and Atom Type (T)
pI	Isoelectric Point
RAM	Random Access Memory
RMSF	Root Mean Square Fluctuation
SDF	Structure Data File
SDS-PAGE	Sodium Dodecyl Sulphate-Polyacrylamide Gel Electrophoresis
SMILES	Simplified Molecular Input Line Entry System
TPSA	Topological Polar Surface Area
UFF	Universal Force Field
UV	Ultraviolet

CHAPTER - 1

INTRODUCTION

CHAPTER- 1
INTRODUCTION

Naegleria fowleri is a single-celled, thermophilic, free-living amoeba belonging to the domain Eukaryota, phylum Heterolobosea, class Eutetramitea, order Acrasida, family Naegleriidae, and genus *Naegleria* (NCBI - NLM, 2026). Commonly found in warm freshwater and soil, it grows by feeding on bacteria in water and soil. Of more than 30 species within the genus *Naegleria*, *N. fowleri* is the only species known to infect humans and is the causative agent of Primary Amoebic Meningoencephalitis (PAM), a rare but rapidly progressing and fatal infection of the central nervous system (Visvesvara et al., 2007).

In 1965, approximately 500 cases were reported globally, with a fatality rate exceeding 95%. West Bengal, Andhra Pradesh, Telangana, Karnataka, Kerala, and Tamil Nadu are among the states in India that have reported approximately 20 cases (Iyer et al., 2024). These states experience higher temperatures, and the majority of cases have been reported during tropical months. Greenhouse gas emissions contribute to rising environmental temperatures, enabling *N. fowleri* to flourish, as it prefers temperatures between 30–46 °C. PAM is associated with age, gender, and recreational activities, with most cases occurring in immunocompetent children and young adults with a median age of 14 years, and approximately 75% of patients being male, typically linked to water-related activities. Infection occurs when contaminated water enters the nasal passages, allowing the amoeba to migrate along the olfactory nerve to the brain, where it causes extensive tissue destruction and inflammation (Iyer et al., 2024; Marri et al., 2025).

The amoeba adheres to the nasal mucosa, penetrates the olfactory epithelium, and migrates along the olfactory nerve through the cribriform plate to reach the olfactory bulbs of the brain, where it proliferates and causes extensive inflammation and tissue destruction (Visvesvara et al., 2007). The incubation period of PAM generally ranges from 1 to 9 days, after which symptoms rapidly develop. *N. fowleri* exhibits three morphological stages in its life cycle: the trophozoite, flagellate, and cyst forms. Among these, the trophozoite stage is the infective and pathogenic form in humans, as it actively feeds, replicates through binary fission, and produces cytolytic molecules and proteolytic enzymes that damage host neural tissue (Visvesvara et al., 2007).

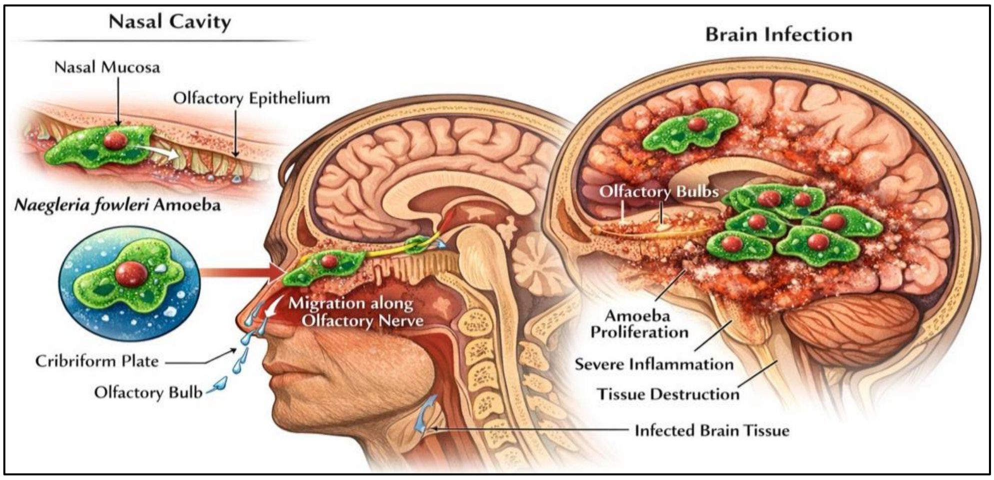

Figure 1: *N. fowleri* pathogenesis, entry through the nasal cavity, migration along the olfactory nerve to the brain, and subsequent proliferation leading to inflammation and extensive neural tissue damage (Iyer et al., 2024)

The flagellate stage is a temporary, non-feeding form that develops under unfavorable environmental conditions to aid motility, while the cyst stage is a dormant survival form that occurs in adverse environments but is not involved in human infection. As the trophozoites multiply within the brain, patients begin to exhibit symptoms resembling acute bacterial meningitis, including severe headache, fever, nausea, vomiting, neck stiffness, altered mental status, seizures, and coma, with disease progression often leading to death within 1–2 weeks after initial exposure if left untreated (Martínez-Castillo et al., 2025).

Based on Güémez and team's observation in the year 2021 (Güémez et al., 2021), management of PAM relies on multidrug therapy due to the rapid progression and extremely high mortality associated with the infection. The antifungal agent Amphotericin B remains the cornerstone of treatment and is typically administered intravenously and, in some cases, intrathecally to achieve adequate concentrations in the central nervous system. Combination therapy is commonly employed to enhance anti-amoebic activity and improve treatment outcomes. Drugs frequently used alongside amphotericin B include Miltefosine, Azithromycin, Fluconazole, and Rifampin (Siddiqui et al., 2024). According to the review, early initiation of treatment is critical for improving clinical outcomes, as PAM progresses rapidly and neurological damage occurs within a short period after symptom onset. Therapy is therefore recommended as soon as PAM is suspected. Despite aggressive treatment, PAM remains associated with extremely high morbidity and mortality, with the majority of patients experiencing severe neurological complications due to extensive brain inflammation and tissue destruction. Only a very small number of survivors have been documented worldwide, and these cases generally involved early diagnosis, prompt initiation of combination therapy, and intensive supportive care. Consequently, although several pharmacological agents have shown activity against *N. fowleri*, the prognosis of PAM remains poor, emphasizing the need for improved therapeutic strategies and early clinical recognition of the disease.

In the present study, Glucokinase of *N. fowleri* (NfGlck) was selected as the target protein due to its fundamental role in the parasite's central carbon metabolism. Glucokinase catalyses the ATP-dependent conversion of glucose to glucose-6-phosphate, constituting the first committed step of glycolysis and a crucial control point in cellular energy production (Milanes et al., 2019). Inhibition of this enzyme may disrupt energy generation, thereby impairing cellular functions necessary for parasite viability. Therefore, NfGlck represents a biologically significant and strategically relevant target for computational drug discovery aimed at developing effective therapeutic interventions against PAM.

Preventive strategies for PAM primarily focus on minimizing exposure to *N. fowleri* in warm freshwater environments. Preventive measures include avoiding swimming or diving in warm freshwater lakes and ponds, particularly during hot weather when amoebal proliferation is increased. The use of nose clips may reduce the risk of nasal exposure. Proper maintenance and chlorination of swimming pools and water facilities are also important preventive strategies (Marri et al., 2025). In addition, the use of sterile, distilled, or boiled water for nasal irrigation practices is strongly recommended to prevent infection. Because PAM is rare and progresses rapidly, prevention mainly relies on environmental management and public awareness.

While vaccination is a well-established approach for infectious disease prevention, its application against *N. fowleri* has been explored only at an experimental level. For instance, a DNA vaccine based on a lentiviral vector expressing the Nfa1 gene has been evaluated in animal models. In vivo studies in mice demonstrated enhanced IgG antibody production along with elevated levels of IL-4 and IFN-γ, indicating activation of both Th1 and Th2 immune responses. Vaccinated animals also exhibited a higher survival rate compared to controls (Siddiqui et al., 2024). However, these findings are limited to preclinical studies, and no vaccine has been approved for human use. Given the low incidence of PAM and the lack of extensive clinical data, vaccine-based approaches remain largely investigational.

Despite advances in understanding the biology and pathogenesis of *N. fowleri*, effective therapeutic options for PAM remain extremely limited. Current treatments rely mainly on repurposed antimicrobial drugs with inconsistent clinical outcomes (Siddiqui et al., 2024). Moreover, the identification of conserved and essential proteins that could serve as potential drug targets in *N. fowleri* has not been extensively explored through multi-omics computational approaches addressing this gap may facilitate the discovery of novel small therapeutic targets for improved treatment strategies against PAM.

CHAPTER - 2

LITERATURE SURVEY

CHAPTER- 2

LITERATURE SURVEY

2.1 Overview of *Naegleri fowleri* and Primary Amoebic Meningoencephalitis (PAM)

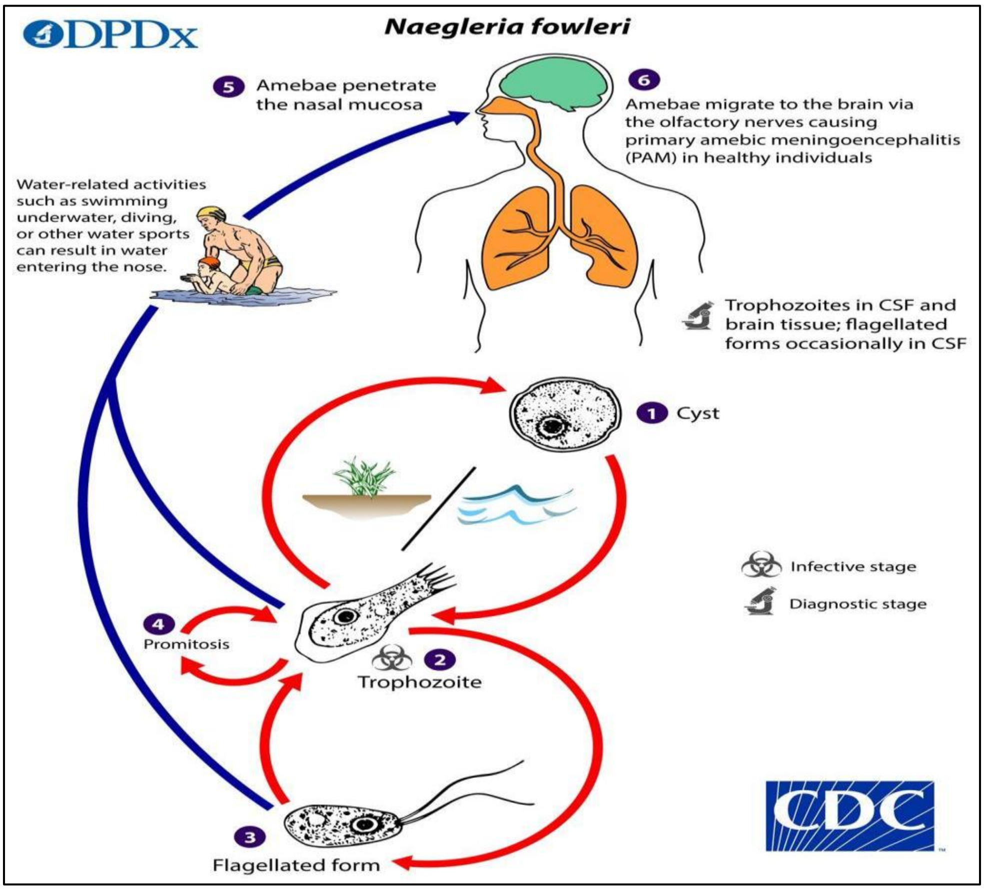

Figure 2: Life cycle and infection pathway of Naegleria fowleri leading to PAM *(CDC - DPDx - Free Living Amebic Infections, 2026).*

Naegleria fowleri is a thermophilic, free-living amoeba commonly found in warm freshwater environments such as lakes, hot springs, and inadequately treated water systems. This protozoan pathogen is the causative agent of Primary Amoebic Meningoencephalitis (PAM), a rare but highly fatal infection of the central nervous system (Visvesvara et al., 2007). The organism enters the human body through the nasal cavity and migrates to the brain via the olfactory nerve, causing rapid inflammation and extensive tissue destruction (Capewell et al., 2015). Although infections are rare, PAM progresses rapidly and has a mortality rate exceeding 95% (CDC, 2025).

2.2 Epidemiology and Clinical Manifestations

Several studies have explored the epidemiology and clinical manifestations of *N. fowleri* infections. PAM cases have been reported worldwide, particularly in warm climates that support amoebic growth (Yoder et al., 2010). Clinical symptoms such as fever, headache, nausea, vomiting, and neurological impairments typically develop within 2–5 days of exposure (Visvesvara et al., 2007). Early diagnosis remains challenging due to symptom similarity with bacterial meningitis, contributing to high fatality rates (Capewell et al., 2015).

2.3 Pathogenic Mechanisms and Virulence Factors

Research into the pathogenic mechanisms of *N. fowleri* has identified several virulence factors. The amoeba uses amoebostomes (food cups) to ingest host cells and secretes proteases and cytolytic enzymes that degrade neural tissue (Marciano-Cabral & Cabral, 2007). These factors facilitate rapid brain tissue destruction and immune evasion. molecular studies have emphasized the role of host–pathogen interactions and identified key proteins and metabolic pathways involved in disease progression (Fritz-Laylin & Fulton, 2016).

2.4 Advances in Molecular Biology and Diagnostic Techniques

Recent advances in molecular biology and omics technologies have enhanced our understanding of *N. fowleri*. Genomic and proteomic analyses have identified potential drug targets and virulence-associated protein (Fritz-Laylin & Fulton, 2016). Advanced diagnostic techniques such as metagenomic next-generation sequencing (mNGS) have improved the speed and accuracy of PAM detection. These approaches are critical for early diagnosis and targeted therapy development.

2.5 Current Treatment Strategies and Limitations

Despite these advancements, effective treatment options remain limited. Amphotericin-B is the primary drug used in PAM treatment due to its ability to disrupt cell membrane integrity (Cope & Ali, 2016). Combination therapy often includes rifampicin, fluconazole, azithromycin, and miltefosine. Rifampicin inhibits RNA synthesis, fluconazole disrupts ergosterol biosynthesis, and azithromycin inhibits protein synthesis, all contributing to anti-amoebic activity (Schuster & Visvesvara, 2004). Miltefosine has shown promise due to its ability to induce apoptosis-like cell death in the amoeba (Cope & Ali, 2016). However, even with aggressive treatment, survival rates remain low.

2.6 Research Gaps and Future Perspective

The present study provides insights into the epidemiology, pathogenicity, and treatment of *N. fowleri*. However, significant gaps remain in understanding its molecular mechanisms and virulence pathways. Further research, particularly involving bioinformatics and molecular characterization of key proteins, is essential for identifying novel therapeutic targets and improving diagnostic strategies. Such advancements could lead to more effective treatments and better medical outcomes.

CHAPTER- 3
THEORETICAL ANALYSIS

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THEORETICAL ANALYSIS

The conceptual framework of this research is centered on the identification of metabolic vulnerabilities in *N. fowleri*, specifically targeting the Glucokinase (NfGlck) enzyme. As an essential catalyst in the glycolytic pathway, NfGlck represents a high-value target for selective toxicity, allowing for the disruption of the pathogen's energy production. The study utilizes molecular docking to simulate the binding orientation and affinity of 2,064 repurposed compounds and their analogs. This process involves a gradient-optimization search to identify the most stable ligand-receptor conformations, where binding affinity is quantified as Gibbs free energy (ΔG) in kcal/mol. To account for the dynamic nature of proteins, CABS-flex 2.0 is employed to calculate Root Mean Square Fluctuation (RMSF), providing a residue-level map of protein flexibility and ensuring the structural persistence of the binding complex.

A critical component of the theoretical evaluation involves assessing the pharmacokinetic feasibility of the lead candidates through ADMET profiling. While Lipinski's Rule of Five is the traditional benchmark for oral bioavailability, the high molecular weight and complexity of potent anti-parasitic agents, such as the reference drug Rifampin, often necessitate a departure from these constraints. In this study, candidates that exceed standard Lipinski parameters are evaluated based on their comparative performance against Rifampin, justifying a shift toward intravenous (IV) administration as the primary delivery route. Furthermore, while traditional blood-brain barrier (BBB) permeability scores may appear unfavorable for some large-molecule leads, the theoretical analysis considers modern pharmaceutical advancements. The potential for utilizing specialized linkers and nano-carrier systems offers a viable solution to facilitate CNS penetration, ensuring that these high-affinity inhibitors can reach the site of infection. Finally, the electronic properties of the leads are validated via Density Functional Theory (DFT) using the B3LYP function. By calculating the HOMO-LUMO energy gap, the study determines the kinetic stability and chemical reactivity of the molecules, ensuring they possess the electronic characteristics required for effective enzymatic inhibition.

CHAPTER- 4

MATERIALS AND METHODOLOGY

CHAPTER- 4

MATERIALS AND METHODOLOGY

All computational analyses in the present investigation were conducted within a comprehensive *in-silico* framework utilizing advanced bioinformatics tools, databases, and molecular modeling platforms. The study was performed on a system powered by an Intel® Core™ Ultra 5 225U processor (12 cores, 14 threads) with 16 GB RAM, ensuring efficient handling of computational workloads. A stable broadband connection enabled uninterrupted access to online bioinformatics resources. The overall workflow encompassed sequence retrieval, proteomic characterization, functional annotation, evolutionary analysis, molecular docking, and pharmacokinetic (ADMET) evaluation of potential inhibitors.

4.1 Retrieval and Validation of NfGlck Protein Sequence

The protein sequence of NfGlck, a pathogenic free-living amoeba responsible for PAM, was retrieved from the UniProt database (https://www.uniprot.org) using the accession ID **A0A384E136**. UniProt is a comprehensive, high-quality, and freely accessible resource for protein sequence and functional annotation data (Apweiler et al., 2004). It integrates data from major bioinformatics institutions including the European Bioinformatics Institute (https://www.ebi.ac.uk), the Swiss Institute of Bioinformatics (https://www.sib.swiss), and the Protein Information Resource (https://pir.georgetown.edu). The UniProt Knowledgebase (UniProtKB) is divided into two main sections: **Swiss-Prot**, which contains manually curated and reviewed protein entries, and **TrEMBL**, which contains computationally annotated entries that are yet to undergo manual curation.

4.2 Proteomic Analysis of NfGlck

Proteomic analysis refers to the large-scale study of proteins, including their physicochemical properties, structural, and functional. Proteins are fundamental biomolecules responsible for a wide range of biological processes such as enzymatic catalysis, molecular transport, signal transduction, and structural integrity. Understanding protein characteristics is essential for identifying drug targets and designing effective therapeutic agents.

In the present study, proteomic analysis of glucokinase was carried out to evaluate its physicochemical properties, structural features, and functional domains.

This analysis provides insights into protein stability, solubility, folding behavior, and interaction potential, which are critical parameters in drug discovery.

4.2.1 Primary Sequence and Physicochemical Property Analysis

The primary structure of a protein refers to the linear sequence of amino acids linked together by peptide bonds. This sequence determines the higher-order structures (secondary, tertiary, and quaternary) and ultimately defines the biological function of the protein. Therefore, primary sequence analysis is a fundamental step in understanding protein characteristics. Primary sequence analysis of glucokinase was performed using the ExPASy server (https://www.expasy.org), which provides access to a wide range of protein analysis tools (Gasteiger et al., 2003).

Among the available tools, the ProtParam tool (https://web.expasy.org/protparam/) was used to compute various physicochemical parameters of the protein. The analysis included determination of molecular weight, theoretical isoelectric point (pI), amino acid composition, atomic composition, extinction coefficient, estimated half-life, instability index, aliphatic index, and grand average of hydropathicity (GRAVY).

4.2.2 Functional Annotation and Conserved Domain Identification

Functional annotation was performed to identify conserved domains, motifs, and functionally important regions within the NfGlck protein sequence. This analysis helps in understanding the structural and functional characteristics of the protein.

The amino acid sequence of NfGlck was retrieved in FASTA format and submitted to the InterPro (https://www.ebi.ac.uk/interpro/) database for domain and family classification. InterPro integrates multiple signature databases, including Pfam, PROSITE, PRINTS, and SMART, enabling the identification of conserved domains, functional sites, and protein family associations (Blum et al., 2021).

The sequence was analysed to identify kinase-related domains, nucleotide-binding regions, and conserved motifs, which are essential for catalytic activity. Special emphasis was given to detecting ATP-binding sites and substrate-binding regions, as these play a critical role in glucokinase function. Domain and motif analysis was further used to predict potential active sites and binding pockets, which are important for downstream molecular docking studies.

4.3 Sequence Similarity Analysis Using BLAST

The Basic Local Alignment Search Tool (BLAST) is a widely used bioinformatics algorithm for comparing biological sequences and identifying regions of local similarity, supporting functional annotation, homology detection, and evolutionary analysis. It employs heuristic approaches to enable rapid and statistically reliable alignments against large databases. BLAST, provided by the National Centre for Biotechnology Information (https://blast.ncbi.nlm.nih.gov/Blast.cgi), offers multiple programs, including BLASTn, BLASTp, BLASTx, tBLASTn, and tBLASTx. In this study, BLASTp was used for protein–protein comparisons to identify homologous sequences and conserved regions (Altschul et al., 1990).

The analysis was conducted sequentially: first against the human protein database to assess potential similarity with host proteins, followed by the PDB to identify structurally characterized homologs, and finally against the non-redundant database with *Naegleria*-specific filtering for evolutionary analysis.

4.3.1 Subtractive Proteomics Analysis Against Homo Sapiens

The BLASTp analysis was performed using the NCBI BLAST server to compare the query protein sequence against *Homo sapiens* (taxid: 9606) protein sequences. The FASTA sequence of NfGlck was used as input, and all algorithm parameters were retained at their default settings. The substitution matrix was set to BLOSUM62, with gap opening and extension penalties of 11 and 1, respectively. The expect threshold (E-value cutoff) was 0.05, ensuring the reporting of statistically significant alignments. A word size of 5 was applied. The maximum number of target sequences was limited to 100. These parameters enable a balance between sensitivity and computational efficiency while maintaining statistical robustness in sequence similarity detection.

4.3.2 Identification of Structural Homologs in PDB

To identify structurally characterized homologs, BLASTp analysis was performed against the PDB, which comprises protein sequences with experimentally resolved three-dimensional structures. The query sequence was aligned against the PDB database using default BLASTp parameters, including the BLOSUM62 substitution matrix, a word size of 5, and gap penalties of 11 for opening and 1 for extension. The expect threshold (E-value) was set to 0.05, and the number of target sequences was limited to 100. The resulting alignments provided structurally resolved homologs of the query protein, from which relevant PDB entries were identified and selected for subsequent structural analysis.

4.3.3 Homology Search Within *Naegleria* Genus Using Non-Redundant Database

BLASTp analysis was performed against the non-redundant (nr) protein database to identify homologous sequences within the *Naegleria* genus. To ensure genus-specific retrieval, the search was restricted by applying an organism filter for *Naegleria* (taxid:5761). The alignment was conducted using standard BLASTp settings, employing the BLOSUM62 substitution matrix with gap opening and extension penalties of 11 and 1, respectively. An expect value threshold of 0.05 was applied to retain statistically significant hits, with a word size of 5 and a maximum of 100 target sequences. The resulting homologous sequences were retrieved and exported in FASTA format for subsequent multiple sequence alignment and phylogenetic analysis.

4.4 Phylogenetic Analysis of NfGlck

Phylogenetic analysis is a fundamental approach to explore evolutionary relationships among biological sequences. It enables the identification of similarities, divergence patterns, and ancestral lineage among proteins from different species. MEGA V12.1 (Molecular Evolutionary Genetics Analysis) (https://www.megasoftware.net/) was employed for multiple sequence alignment (MSA) and the reconstruction of phylogenetic relationships. This software suite provides a biologist-centric interface for estimating evolutionary distances and testing evolutionary hypotheses through various statistical methods (Kumar et al., 2024).

4.4.1 Phylogenetic Tree Construction using MEGA V12.1 Software

The FASTA sequences of selected *Naegleria* species were imported into MEGA V12.1 software using the Alignment Explorer module, where a new protein alignment dataset was generated. All homologous sequences were organized within the workspace to facilitate structured analysis. The dataset was subsequently converted into MEGA V12.1 format (.meg) to ensure compatibility with downstream phylogenetic tools. Multiple sequence alignment (MSA) was performed using the ClustalW algorithm implemented in MEGA V12.1, with default parameters retained. This step enabled precise alignment of homologous amino acid residues, forming a robust basis for subsequent evolutionary and phylogenetic inference.

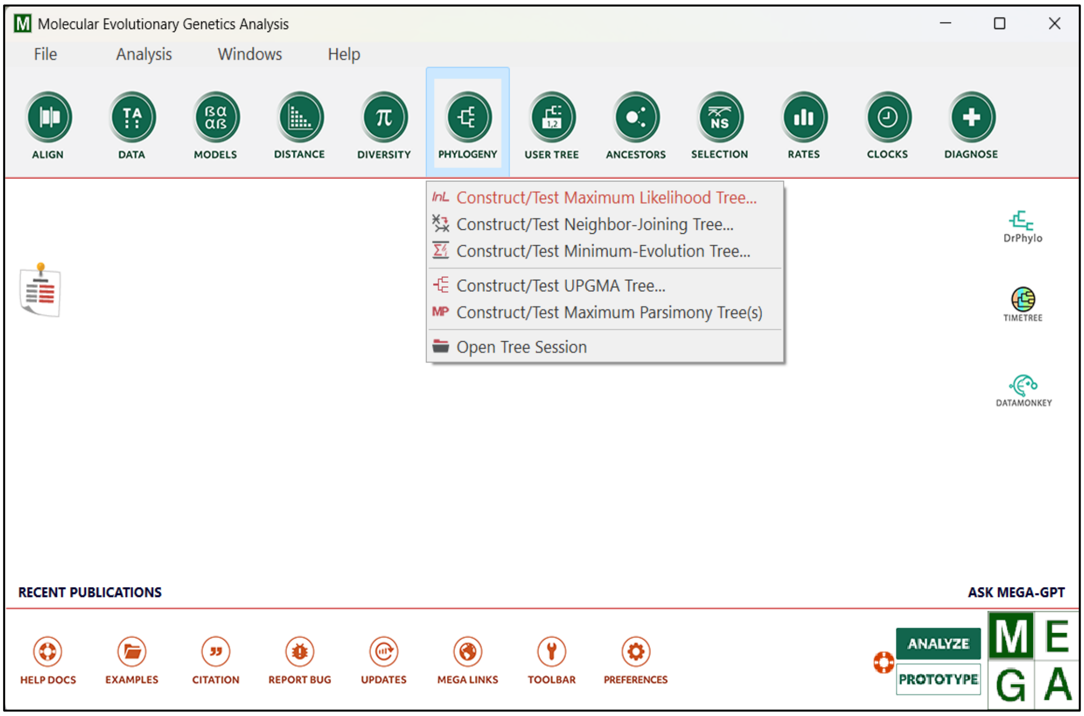


Figure 3: MEGA V12.1 software interface.

"Phylogeny" dropdown menu (Figure 3) and available tree construction methods (Maximum Likelihood, Neighbour-Joining, etc.)

The aligned sequences were subsequently subjected to phylogenetic tree construction using the Maximum Likelihood (ML) method. The phylogenetic tree was visualized using the Tree Explorer module in MEGA V12.1, where branch lengths, topology, and sequence labels were carefully examined. The tree was then formatted appropriately and exported as an image for inclusion in the thesis.

4.5 Docking

4.5.1 Protein Preparation

The PDB (https://www.rcsb.org/) is a comprehensive repository of experimentally determined three-dimensional structures of biomolecules, including proteins and nucleic acids, resolved using techniques such as X-ray crystallography, NMR spectroscopy, and cryo-electron microscopy. It serves as a critical resource for structural biology, enabling detailed analysis of protein architecture, functional sites, and molecular interactions (Burley et al., 2019).

In the present study, the three-dimensional structure of NfGlck from *Naegleria fowleri* was retrieved from the PDB (PDB ID: 6DA0). This structure was selected based on its high-resolution X-ray crystallographic data (2.20 Å) and the availability of detailed experimental validation metrics, indicating good model quality and reliability.

The structure also contains a co-crystallized ligand, ANP, which facilitated precise identification of the active site region (Diffraction Project Datasets 6da0, 2026; Siddiqui et al., 2024).

The structure was downloaded in PDB format and subsequently utilized for protein–ligand interaction analysis and molecular docking studies, where the experimentally validated active site information contributed to improved accuracy in binding site prediction and docking outcomes.

4.5.2 Protein-Ligand Interaction Analysis

PDBsum (https://www.ebi.ac.uk/pdbsum/) was used to obtain a comprehensive pictorial summary and structural analysis of the protein targets. It acts as an "atlas" for each structure in the PDB, providing an at-a-glance overview of its biological molecules, secondary structure, and molecular interactions (Laskowski et al., 2017).

The protein–ligand interaction analysis was performed using the PDBsum server based on the crystal structure of NfGlck (PDB ID: 6DA0) obtained from the PDB. The three-dimensional structure of glucokinase complexed with the co-crystal ligand ANP (Adenosine Nucleotide Analog) was analyzed to identify intermolecular interactions within the active site. The PDBsum server was used to determine hydrogen bonds and non-bonded interactions between the protein and ligand. Hydrogen bonds were identified based on donor–acceptor distances typically ranging between 2.5 Å and 3.5 Å, while non-bonded contacts were considered within a distance of less than 4.0 Å. The interacting residues, atom types, and bond distances were recorded for detailed analysis. The identified interaction residues were further used to define the active site region and served as a reference for molecular docking studies.

4.5.3 Selection and Preparation of Ligands for HTVS Docking

The identification of potential ligands for molecular docking was conducted using a systematic approach that integrates literature-based selection with structure-based similarity analysis. This dual strategy ensured the inclusion of both experimentally validated drugs and structurally relevant analogs, providing a reliable framework for computational screening against NfGlck. Within this framework, the ANP ligand was selected as a positive control compound due to its direct involvement in enzyme catalytic activity and its established binding interactions within the active site.

A set of repurposed drug candidates reported for the treatment of *N. fowleri* infections was selected based on literature evidence (Dai et al., 2025). Each selected compound was individually queried using its respective PubChem Compound ID (CID). These molecules—Amphotericin B (CID: 5280965), Azithromycin (CID: 447043), Rifampin (CID: 135398735), Miltefosine (CID: 3599), and Fluconazole (CID: 3365) were established as literature-based reference ligands owing to their therapeutic relevance and documented anti-fungal and antibiotic drugs.

4.5.4 Similarity-Based Ligand Identification and Data Retrieval Using PubChem

HTVS was performed following ligand selection, structural similarity searches were performed using the PubChem database(https://pubchem.ncbi.nlm.nih.gov). PubChem is a widely used, publicly accessible repository that provides curated information on chemical compounds, including their molecular structures, physicochemical properties, and biological activities. One of the major advantages of PubChem is the availability of standardized chemical formats such as SDF and MOL2, which are directly compatible with molecular docking and simulation software. Furthermore, its advanced search capabilities allow identification of structurally related compounds through similarity-based algorithms, making it highly suitable for virtual screening workflows (Kim et al., 2025).

The similarity search was conducted using a two-dimensional (2D) molecular fingerprint-based approach. Structural similarity between compounds was quantified using the Tanimoto coefficient, a widely accepted metric in cheminformatics that measures the degree of overlap between molecular fingerprints on a scale ranging from 0 (no similarity) to 1 (identical structures). To ensure high precision in ligand selection, a stringent similarity threshold of 99% was applied instead of the default 90%, allowing retrieval of only near-identical analogs. The identified compounds were subsequently downloaded in SDF format for use in high-throughput virtual screening.

4.6 Computational Docking

Protein, ligand preparation and high-throughput virtual screening were performed using PyRx V0.8, an open-source computational platform extensively used in structure-based drug discovery. PyRx V0.8 integrates Open Babel for efficient molecular file format conversion and supports energy minimization using the Universal Force Field (UFF). Additionally, it incorporates AutoDock Vina for molecular docking, enabling a streamlined workflow within a single interface. The platform was selected due to its user-friendly design, ability to handle large ligand datasets, and reliable performance in virtual screening studies (Dallakyan & Olson, 2015; O'Boyle et al., 2011; Ram et al., 2022; Trott & Olson, 2010).

4.6.1 Receptor Preparation and Optimization

The protein preparation process was performed to ensure that the target structure was suitable for accurate molecular docking analysis. The experimentally resolved protein structure, with all water molecules removed, was imported into the PyRx V0.8 workspace for preprocessing. Within PyRx V0.8, receptor preparation is carried out through automated routines that standardize the protein structure for docking compatibility.

During preprocessing, missing hydrogen atoms were added to the protein structure to correctly represent protonation states and enable accurate hydrogen bonding interactions. Partial atomic charges were assigned to the receptor, facilitating proper electrostatic interaction calculations during docking. The software also ensures appropriate atom type assignment and structural consistency required for downstream analysis.

In the background, PyRx V0.8 performs essential preprocessing steps including correction of structural geometry, assignment of bonding parameters, and preparation of the receptor in a format compatible with docking engines. The processed protein is subsequently converted into PDBQT format, which incorporates atom types and charge information necessary for docking using AutoDock Vina. Proper receptor preparation is critical for reliable docking results, as it ensures accurate representation of the protein's chemical environment and facilitates precise prediction of protein–ligand interactions within the active site.

4.6.2 Ligand Preprocessing and Geometry Optimization

The ligand preparation process was carried out systematically to ensure that all compounds were structurally optimized and suitable for docking analysis. Ligands obtained in SDF format from the PubChem database were imported into the PyRx V0.8 workspace, where they were converted into three-dimensional conformations using Open Babel for standardized structural handling. Following conversion, all ligands underwent geometry optimization and energy minimization using the Universal Force Field (UFF) implemented within PyRx V0.8 (Ram et al., 2022), enabling the generation of energetically stable conformations by reducing steric strain and optimizing bond geometry. Polar hydrogens were added, and partial atomic charges were assigned to ensure accurate representation of electrostatic interactions essential for docking calculations. In addition, rotatable bonds were defined to allow conformational flexibility during the docking process.

The optimized ligands were then converted into PDBQT format, incorporating atom types, charge information, and torsional degrees of freedom required for compatibility with AutoDock Vina (O'Boyle et al., 2011). Proper ligand preparation is critical for reliable docking outcomes, as it ensures that the molecular structures closely approximate biologically relevant conformations, thereby improving the accuracy of predicted binding interactions.

4.6.3 Docking-Based Virtual Screening Using AutoDock Vina

High-throughput virtual screening was performed using AutoDock Vina integrated within PyRx V0.8 to evaluate the binding affinity of the prepared ligands against glucokinase. The docking grid was defined based on the position of the co-crystallized ligand ANP, targeting the biologically relevant ATP-binding pocket. The grid box was centred at coordinates X = 76.07, Y = 44.55, and Z = 7.72, with dimensions of 9.61 Å × 13.90 Å × 13.61 Å, ensuring that all ligands were docked within the active site region.

white box (Figure 4) represents the defined search space of the target protein, while pink spheres indicate ligand positions within the docking region. Docking simulations were conducted using the standard parameters of AutoDock Vina, including an exhaustiveness value of 8 to ensure adequate conformational sampling and generation of up to 9 binding modes allowing the generation of multiple binding conformations for each ligand. The best binding pose for each compound was selected based on the lowest predicted binding energy, which indicates the most stable interaction with the target protein (Trott & Olson, 2010). The resulting docked complexes were subsequently analyzed to evaluate binding affinities and interaction patterns, with particular emphasis on hydrogen bonding and interactions with key active site residues. This comprehensive analysis enabled the identification of ligands with

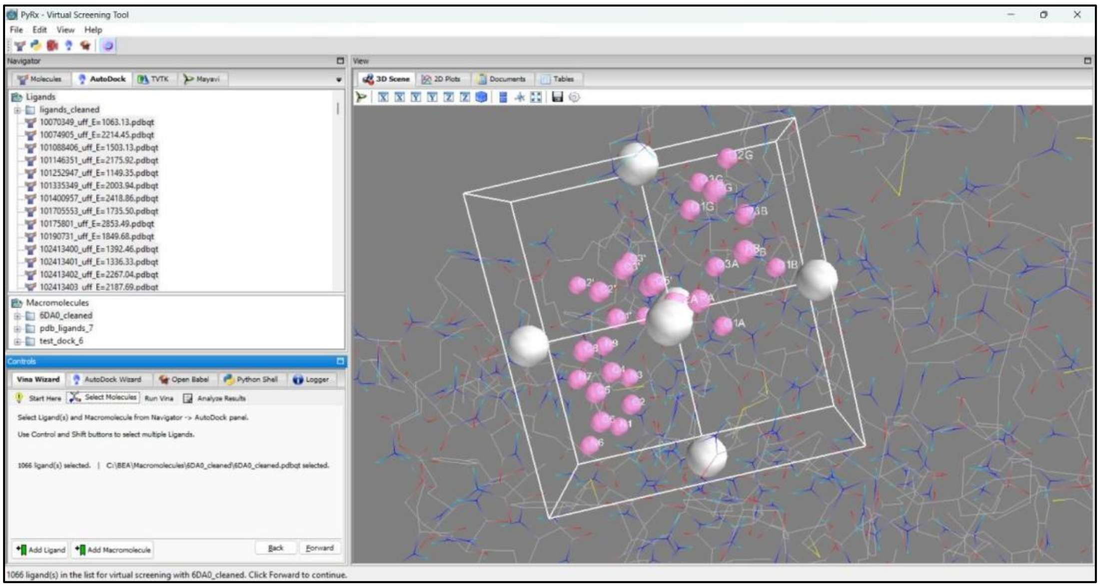


Figure 4: Docking Grid Box Configuration in PyRx V0.8.

strong binding potential and suitability for further structure-based drug design and validation studies.

4.7 Pharmacokinetic Analysis

In-silico pharmacokinetic and toxicity profiling of the top-ranked ligands was performed using ADMETlab 3.0 (https://admetlab3.scbdd.com/). It is an advanced web-based platform that integrates machine learning and curated datasets to predict absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties of small molecules. This tool enables rapid and reliable evaluation of drug-likeness and safety profiles, facilitating early-stage screening in drug discovery pipelines (Fu et al., 2024).

The SMILES representations of selected ligands, identified based on docking performance, were used as input for ADMET analysis. The platform computes a wide range of physicochemical, pharmacokinetic, and toxicity-related descriptors using validated predictive models. These predictions provide insights into key properties such as oral bioavailability, membrane permeability, metabolic stability, and potential toxicity risks. The obtained ADMET profiles were subsequently analyzed to assess the drug-likeness and pharmacological suitability of the selected compounds.

4.8 Protein Flexibility Analysis Using Cabs-Flex 2.0

4.8.1 Simulation Setup and System Preparation

To evaluate the dynamic stability and structural flexibility of protein–ligand complexes obtained after molecular docking, simulations were performed using the CABS-flex 2.0 web server (http://biocomp.chem.uw.edu.pl/CABSflex2), a reliable and computationally efficient coarse-grained modeling tool for protein dynamics analysis. This platform simplifies protein representation while preserving essential structural and interaction features required for accurate prediction of conformational behaviour (Kuriata et al., 2018).

In the present study, the best docking poses of all selected systems, including the native ANP complex (Native), re-docked ANP complex (33113_ANP), reference drug rifampin complex, and shortlisted lead compounds, were exported in PDB format and submitted to the CABS-flex 2.0 server. All simulations were performed at 500 cycles to ensure consistency and reproducibility across all protein–ligand systems. The simulations generated an ensemble of conformations representing the intrinsic flexibility of the protein structure under near-physiological conditions.

4.8.2 MD simulation analysis

The global flexibility of the protein structure was analyzed at the chain level using Root Mean Square Fluctuation (RMSF), which measures the average positional deviation of each amino acid residue during the simulation trajectory. RMSF data obtained from the CABS-flex output were exported and compiled in Microsoft Excel for systematic analysis.

These data were used to generate dot-scatter line plots (RMSF vs residue number), enabling visualization of residue-wise fluctuations across the entire protein sequence. This analysis facilitated validation of the re-docked ANP complex by comparison with the native ANP system and enabled comparative assessment of all ligand-bound complexes at the global structural level.

4.8.3 Binding Site Residue-Level Analysis

To investigate ligand-induced effects at the active site, RMSF values corresponding to key binding pocket residues (GLY56, ALA57, THR58, ASN59, ALA116, ASN128, ASP156, ALA204, GLY206, THR207, GLY208, GLY210, GLY340, ASP341, and ASN342) were selectively extracted from the complete dataset. These residues represent critical regions involved in ligand binding and structural stability. The extracted RMSF values were organized in Excel and used to construct grouped bar charts with numerical annotations, allowing direct comparison of residue-specific flexibility across all protein–ligand complexes. This approach enabled detailed evaluation of local structural dynamics and identification of flexible hotspots and stable core regions within the binding pocket.

4.9 Frontier Molecular Orbital (FMO) Analysis Using DFT

The electronic properties of the selected top ligands were investigated through Frontier Molecular Orbital (FMO) analysis employing Density Functional Theory (DFT)-based computational approaches. FMO analysis provides valuable insights into molecular reactivity, stability, and charge transfer behaviour by evaluating the energies of frontier orbitals. DFT, a widely adopted quantum mechanical method, offers an optimal balance between computational efficiency and accuracy, making it suitable for studying the electronic structure of biologically relevant molecules.

The computational workflow was implemented using a Python-based pipeline integrating scientific libraries such as *NumPy* for numerical computations, *Pandas* for data handling, and *Matplotlib* for visualization, along with cheminformatics tools for molecular processing and descriptor calculation. Initial ligand structures were generated from SMILES notation and subjected to geometry optimization using the MMFF94 force field to obtain energetically stable conformations. The optimized structures were subsequently used for DFT calculations employing the B3LYP exchange–correlation functional with the STO-3G basis set. In instances where convergence was not achieved, calculations were repeated using the PBE functional to ensure computational robustness.

The code utilized in this study was adapted from a publicly available Kaggle repository, and the complete implementation is provided in Appendix, enabling reproducibility and systematic evaluation of quantum chemical parameters for the shortlisted ligands. Additionally, HOMO-LUMO 2d slice images were generated for visualization purposes.

4.9.1 HOMO-LUMO and Energy Gap

The Highest Occupied Molecular Orbital (HOMO) and Lowest Unoccupied Molecular Orbital (LUMO) describe a molecule’s ability to donate and accept electrons, respectively. The energy difference between them, known as the HOMO–LUMO gap (ΔE), is a key indicator of molecular stability and reactivity. A smaller ΔE suggests higher reactivity and lower stability, which can enhance interactions with biological targets, while a larger ΔE indicates greater stability and reduced reactivity. These properties play a crucial role in determining ligand behaviour and binding potential (Mahmoudi & Kim, 2026). The HOMO–LUMO energy gap (ΔE) was computed as (Domingo et al., 2016).

$$Energy\ Gap\ (\Delta E) = E_{LUMO} - E_{HOMO} ----(1)$$

4.9.2 Global Reactivity Descriptors: Hardness and Softness

Chemical hardness (η) and softness (S) are key reactivity descriptors derived from HOMO and LUMO energies. Hardness is defined as half of the HOMO–LUMO gap, while softness is its inverse. Molecules with higher hardness (larger energy gap) are generally more stable and less reactive, whereas softer molecules (smaller gap) are more polarizable and chemically reactive. In ligand–protein interactions, softer ligands tend to better adapt to the electronic environment of the binding site, improving interaction potential (Domingo et al., 2016; Kaya & Putz, 2022; Miar et al., 2021).

$$\text{Chemical hardness } (\eta) = \frac{E_{LUMO} - E_{HOMO}}{2} ----(2)$$

$$\text{Chemical softness } (S) = \frac{1}{\eta} ---------- (3)$$

4.9.3 Electronegativity and Electrophilicity

Electronegativity (χ) and electrophilicity index (ω) are important descriptors that reflect a molecule's tendency to attract and accept electrons. Electronegativity indicates electron-attracting ability, while electrophilicity measures the stabilization gained upon electron acceptance. Higher electrophilicity enhances interactions with nucleophilic residues in protein binding sites, while electronegativity influences charge distribution and binding orientation. Together, these parameters are useful for evaluating ligand reactivity, efficiency, and binding potential (Domingo et al., 2016; Miar et al., 2021).

$$\text{Electronegativity } (\chi) = -\frac{E_{HOMO} + E_{LUMO}}{2} -----(4)$$

$$\text{Electrophilicity } (\omega) = \frac{\chi^2}{2\eta} -----------(5)$$

From the above equations 1,2,3, 4 and 5:

- E_{HOMO}: Energy of the Highest Occupied Molecular Orbital
- E_{LUMO}: Energy of the Lowest Unoccupied Molecular Orbital
- ΔE: HOMO–LUMO energy gap

- η: Chemical hardness
- S: Chemical softness
- χ: Electronegativity
- ω: Electrophilicity index

CHAPTER- 5

RESULTS

CHAPTER- 5
RESULTS

5.1 Retrieval of Target Protein Sequence

The protein sequence retrieved from the UniProt database (Table 1) corresponds to glucokinase (EC: 2.7.1.2) from *N. fowleri*, a pathogenic amoeba commonly referred to as the brain-eating amoeba. The protein is associated with the primary Uniprot ID A0A384E136 and is classified under the taxonomic identifier 5763 (NCBI). Taxonomic lineage analysis revealed that the organism belongs to the domain Eukaryota, within the class Heterolobosea and family Vahlkampfiidae.

Table 1: FASTA Sequence of glucokinase (EC: 2.7.1.2) from *N. fowleri.*

>tr|A0A384E136|A0A384E136_NAEFO Glucokinase OS=Naegleria fowleri OX=5763
MPERMISSSSRSSMSSSTSSEEPLVIDEDFINQVEKHYRKSLKNIPFQYIVGVDVGAT NTRIAIQFIINEDQDDEVFMTKFPCNTSTHLANYLAAYGKAMVKAVGKGSAAGSIA LAGPVTGDKVRITNYKEHDQEFFYSQLPDTLFPASKNTFLNDLEASCYGIINVGTN NRLHEFFCPIDALNNYATSQTVRLSDTSEYAVLAMGTGLGTGLIVGSAGGKFNVIP LEAGHVHIATPGVNSEHFKEERERIEFLSQKIYGGAYPIEYEDICSGRGLEFCYEFEI RNDPNAVRKTASQIAESYSTDTYARQAMITHYRYLMKAAQNIAVLIPTCRGVFFA GDNQVFNEDFFKEHLSILQKELFQTHQKKHWLTDLKPYRQMKEYNFNVKGCLQK ARELAQLGSTTMVDEKPKAGLKDVLSVAIPTAFFSVLSFFTAKNLYEK

5.2 PROTEOMIC ANALYSIS

5.2.1 Primary Sequence Analysis

The physicochemical properties of NfGlck were analysed using the ExPASy server via the ProtParam tool. The computed parameters are summarized in Table 2.

Table 2: The physicochemical parameters.

Parameter	Values
Instability index	27.51 (Stable)
Aliphatic index	78.32
GRAVY	-0.278
Extinction coefficient (oxidized)	35,675 M^{-1} cm^{-1}
Extinction coefficient (reduced)	35,300 M^{-1} cm^{-1}
Estimated half-life (mammalian cells)	30 hours

Table 3: Distribution of amino acids in the protein sequence, expressed as residue count and percentage, calculated using ProtParam.

Amino Acid	Count	Percentage (%)
Alanine (A)	36	8.2%
Arginine (R)	17	3.9%
Asparagine (N)	24	5.4%
Aspartic acid (D)	20	4.5%
Cysteine (C)	7	1.6%
Glutamine (Q)	18	4.1%
Glutamic acid (E)	31	7.0%
Glycine (G)	28	6.3%
Histidine (H)	11	2.5%
Isoleucine (I)	28	6.3%
Leucine (L)	32	7.3%
Lysine (K)	28	6.3%
Methionine (M)	10	2.3%
Phenylalanine (F)	27	6.1%
Proline (P)	16	3.6%
Serine (S)	33	7.5%
Threonine (T)	28	6.3%
Tryptophan (W)	1	0.2%
Tyrosine (Y)	20	4.5%
Valine (V)	26	5.9%

The NfGlck consists of 441 amino acids with a calculated molecular weight of 49.36 kDa. The amino acid composition analysis revealed that residues such as alanine (8.2%), leucine (7.3%), serine (7.5%), and glutamic acid (7.0%) are a closed distribution of hydrophobic and hydrophilic residues. The total number of negatively charged residues (51) is slightly higher than positively charged residues (45), further supporting the acidic nature of the protein. The atomic composition of the protein was determined as $C_{2207}H_{3413}N_{585}O_{667}S_{17}$, with a total of 6889 atoms, reflecting the complex molecular structure of the enzyme. The extinction coefficient was calculated to be 35,675 M^{-1} cm^{-1} (oxidized) and 35,300 M^{-1} cm^{-1} (reduced), indicating that the protein can be effectively detected and quantified using UV spectrophotometry at 280 nm.

The estimated half-life of the NfGlck was found to be 30 hours in mammalian reticulocytes, greater than 20 hours in yeast, and greater than 10 hours in *Escherichia* *coli*, suggesting that the protein exhibits considerable stability across different biological systems. The instability index was calculated to be 27.51, classifying the protein as stable under in vitro conditions. The aliphatic index of 78.32 indicates good thermal stability, while the GRAVY value of -0.278 suggests that the protein is hydrophilic in nature.

In summary, physicochemical analysis shows that glucokinase is a stable, moderately sized enzyme with favourable properties for ligand interaction. Its slightly acidic pI indicates a negative charge at physiological pH, influencing electrostatic binding. The low instability index and high aliphatic index confirm structural and thermal stability, while the negative GRAVY value highlights hydrophilicity, supporting cytosolic function. Charged and aromatic residues further suggest potential sites for ionic, hydrogen bonding, and π-interactions during docking studies.

5.2.2 Functional annotation and domain analysis

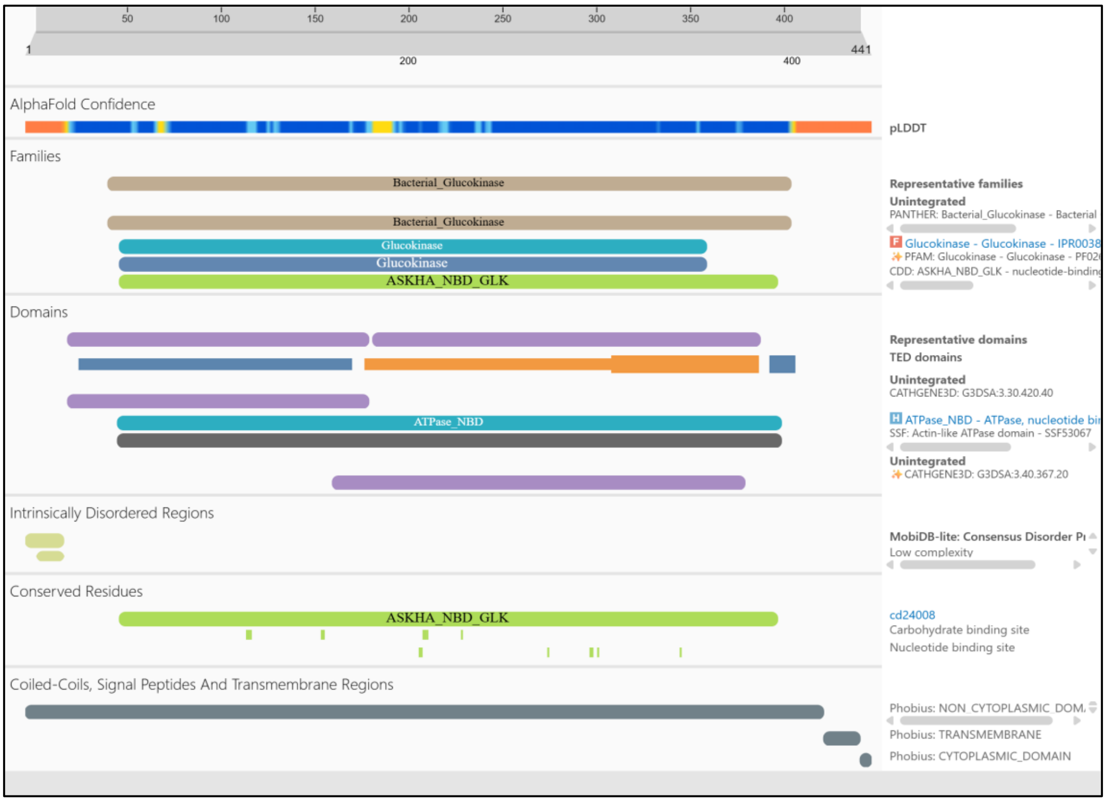


Figure 5: Domain architecture of glucokinase predicted by InterPro.

InterPro analysis (Figure 5) confirms that NfGlck belongs to the glucokinase family, with a clearly defined conserved glucokinase domain spanning the majority of the sequence. This classification supports its role in ATP-dependent glucose phosphorylation, a key step in glycolysis. The figure further shows a prominent ATP-binding nucleotide-binding domain (NBD), along with conserved residues mapped within this region, indicating a preserved catalytic core required for substrate binding and phosphoryl transfer. Additional annotations of carbohydrate and nucleotide binding sites reinforce its functional specificity. Together, the conserved domain architecture and family assignment validate NfGlck as a structurally and functionally reliable target, supporting its suitability for downstream inhibitor design.

5.3 BLAST analysis

5.3.1 Subtractive Proteomics Analysis

The BLASTp analysis of glucokinase (441 amino acids) against the human protein database (taxid:9606) showed no significant similarity, indicating the absence of homologous proteins in humans. The lack of similarity reduces the possibility of cross-reactivity and off-target effects, thereby supporting its suitability as a selective therapeutic target.

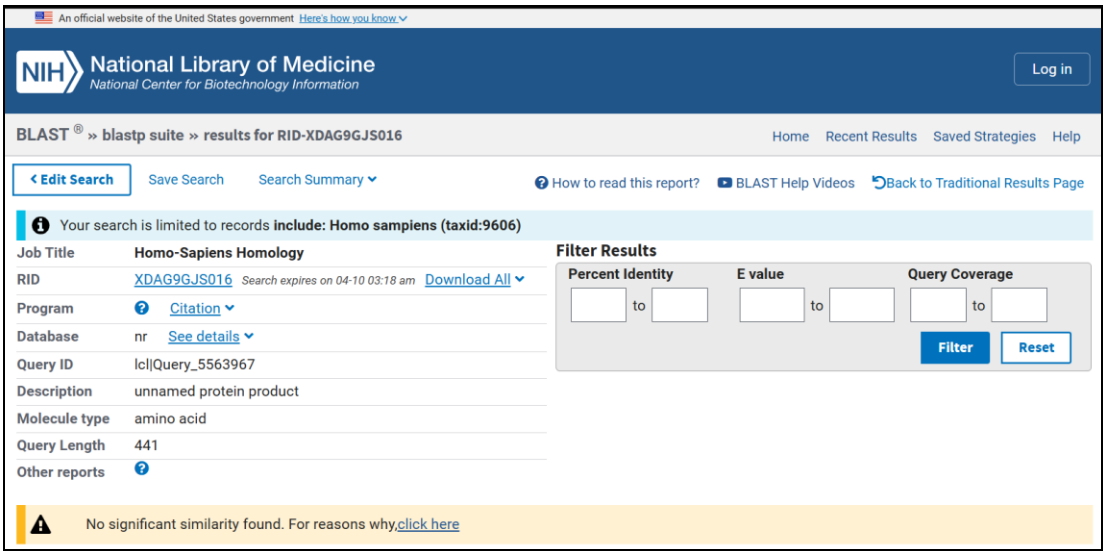
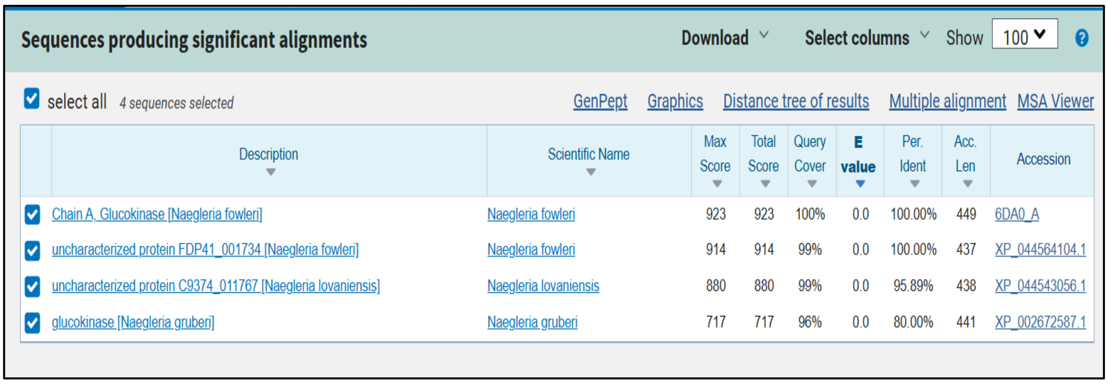


Figure 6: BLASTp analysis showing non-homology of NfGlck with *Homo sapiens.*

5.3.2 Evolutionary Analysis within *Naegleria*



Sequences producing significant alignments | Download | Select columns | Show 100

select all 4 sequences selected | GenPept | Graphics | Distance tree of results | Multiple alignment | MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Chain A, Glucokinase [Naegleria fowleri]	Naegleria fowleri	923	923	100%	0.0	100.00%	449	6DA0_A
uncharacterized protein FDP41_001734 [Naegleria fowleri]	Naegleria fowleri	914	914	99%	0.0	100.00%	437	XP_044564104.1
uncharacterized protein C9374_011767 [Naegleria lovaniensis]	Naegleria lovaniensis	880	880	99%	0.0	95.89%	438	XP_044543056.1
glucokinase [Naegleria gruberi]	Naegleria gruberi	717	717	96%	0.0	80.00%	441	XP_002672587.1

Figure 7: The figure represents BLASTp alignment results of NfGlck across Naegleria species.

BLASTp analysis against the non-redundant protein database, filtered for the *Naegleria* genus, revealed several significant homologs. The selected sequences showed high query coverage (96–100%), E-values (<0.01), and high percentage identity (80–100%), indicating strong sequence conservation. BLASTp hits included *Naegleria fowleri* (100% identity, 100% query coverage), *Naegleria lovaniensis* (95.89% identity, 99% query coverage), and *Naegleria gruberi* (80% identity, 96% query coverage). These BLAST results revealed, NfGlck is highly conserved within the *Naegleria* genus, suggesting its essential role in metabolic processes such as glucose phosphorylation in *Naegleria* species.

5.4 Phylogenetic Analysis

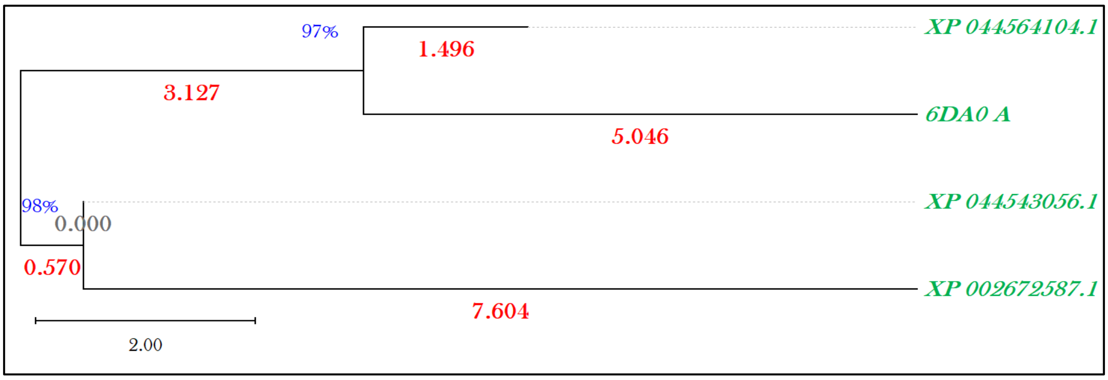


Figure 8: Phylogenetic tree constructed using the Maximum Likelihood method by MEGA V12.1.

The evolutionary relationships among NfGlck sequences from *Naegleria* species (Figure 8). Coverage percentages are shown in blue, branch lengths in red, and UniProt/PDB sequence IDs in green, representing proteins from different *Naegleria* species. The scale bar (2.00) indicates amino acid substitutions per site. The phylogenetic tree (Figure 8), constructed using the Maximum Likelihood method in MEGA V12.1.

The evolutionary relationships among NfGlck sequences from different *Naegleria* species. The tree shows clear clustering patterns, indicating varying levels of sequence conservation and divergence. The sequences 6DA0_A and XP_044564104.1 from *N. fowleri* cluster closely corresponds with strong bootstrap support (~97%). The sequence XP_044543056.1 from *N. lovaniensis* forms a nearby but distinct relevance with high bootstrap support (~98%). In contrast, XP_002672587.1 from *N. gruberi* shows the greatest divergence, as indicated by its longer branch length (7.6).

5.5 Three-Dimensional Structure Retrieval and Evaluation Using PDBsum

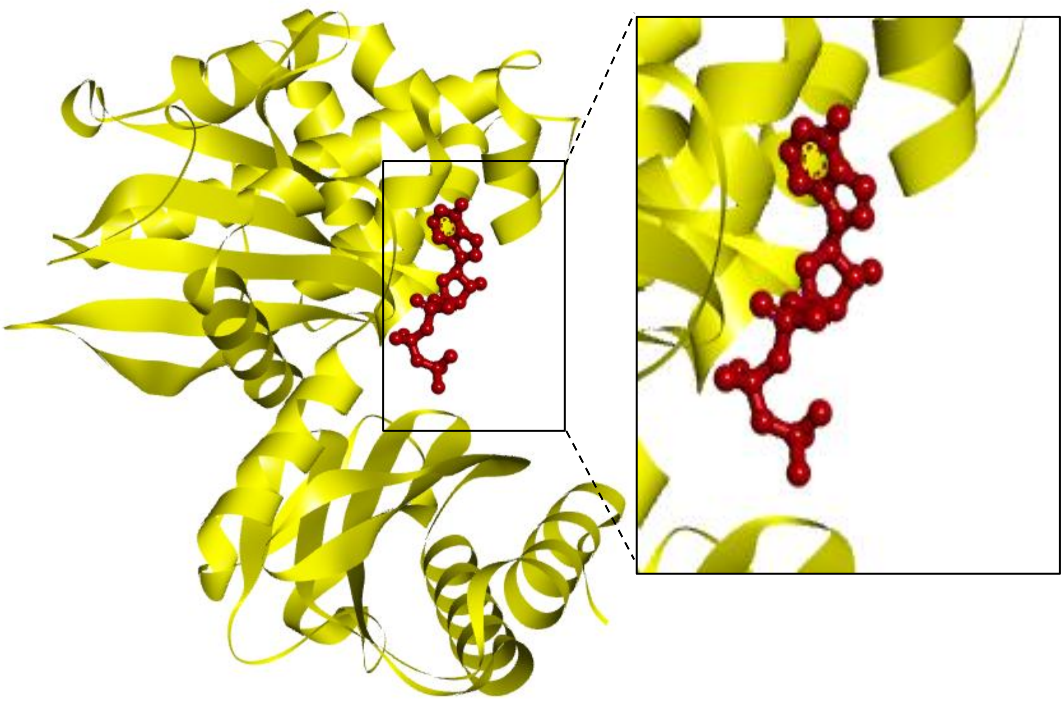


Figure 9: Ligand–protein interaction of ANP and NfGlck (PDB ID: 6DA0).

The retrieved structure (PDB ID: 6DA0) represents a high-quality, experimentally validated model of NfGlck. The presence of co-crystallized ligands (ANP) at residue 501confirms the location of the active site (Figure 9), making it suitable for further interaction analysis and molecular docking studies. The structure of NfGlck was determined using X-ray diffraction at a resolution of 2.20 Å, indicating a high level of structural accuracy and reliability. The protein is represented by chain A and consists of a total sequence length of 449 amino acids, of which 378 residues were successfully modeled in the resolved structure. It exists as a monomeric protein with an approximate molecular weight of 51.13 kDa, making it suitable for detailed structural and functional analysis.

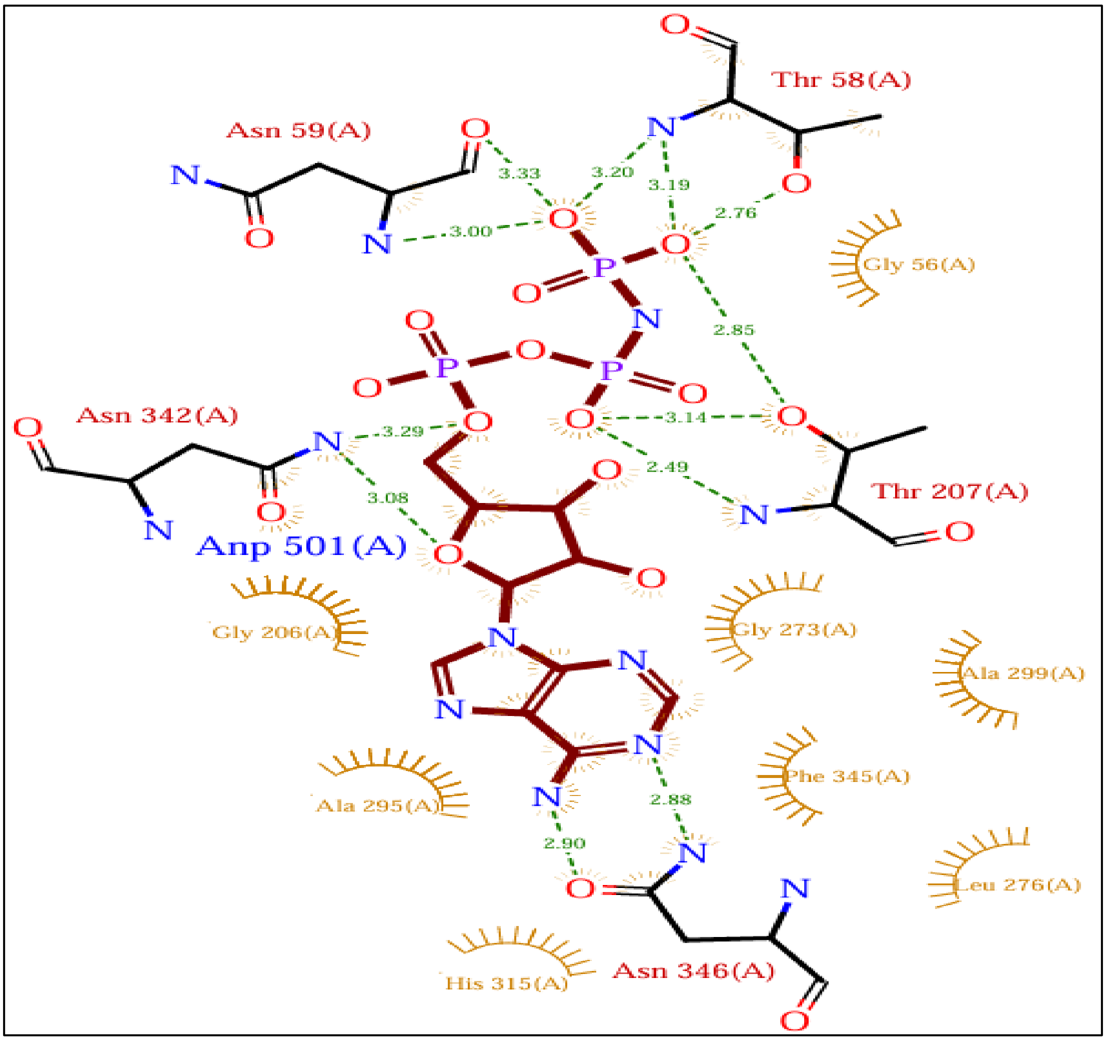


Figure 10: Molecular interaction between co-crystallized ligand ANP with NfGlck.

Interaction of co-crystallized ligand ANP 501 with key residues in the protein active site. Hydrogen bonds (green lines) and van der Waals contacts (red arcs) define the structural basis of ligand–protein recognition, offering insights relevant to drug design.

Analysis of the ANP-bound structure (PDB ID: 6DA0) using PDBsum (Figure 10) reveals a well-organized interaction network within the ATP-binding pocket of NfGlck. ANP is centrally positioned and stabilized by multiple hydrogen bonds with distances ranging from ~2.4 to 3.3 Å, indicating strong binding affinity. Critical residues include Thr58, Asn59, Thr207, Asn342, and Asn346. Thr58 and Thr207 primarily interact with the phosphate groups, while Asn342 and Asn346 contribute to stabilization of the ribose and adenine moieties. In addition, surrounding residues such as Gly56, Gly206, Gly273, Ala295, Ala299, Leu276, His315, and Phe345 form extensive van der Waals interactions, creating a compact and well-defined binding environment that further enhances ligand stability.

5.6 High-Throughput Virtual Screening (HTVS)

The distribution of these analogs revealed significant variation in the chemical "neighborhoods" of the selected templates. Amphotericin B provided the most extensive representation with 964 similar compounds, followed by Azithromycin and Rifampin, which contributed 426 and 415 respectively.

In contrast, the co-crystallized reference ANP demonstrated a moderate availability of 202 analogs, while Miltefosine and Fluconazole exhibited the most restricted structural diversity, returning only 30 and 27 compounds under the applied 99% similarity constraints. The stand-alone dataset contains a total of 2,064 compounds established for the subsequent molecular docking, binding affinity evaluation, and ADMET profiling targeting NfGlck.

5.7 Ligand Conversion and Energy Minimization

Out of the 2064 retrieved ligands, 2018 compounds were successfully prepared for docking after format conversion (SDF to PDBQT) and energy minimization using PyRx V0.8 with the Universal Force Field (UFF). All ligands were verified for structural integrity, protonation, and absence of duplicates. The resulting docking-ready library represents a chemically diverse and most favorable dataset suitable for docking against NfGlck.

5.8 Evaluation of Docking Scores

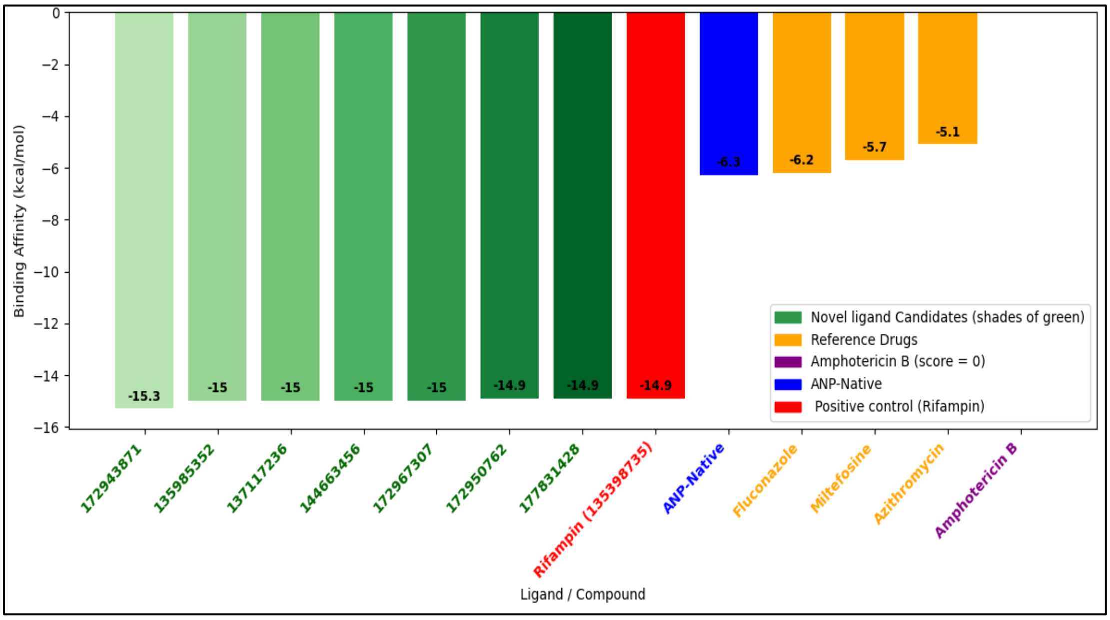


Figure 11: Binding affinities of novel ligands after ADMET screening compared with controls.

Binding affinities of the screened novel ligands in comparison with the control compounds (Figure 11). The novel ligands (represented in shades of green) demonstrate strong binding affinities ranging from –14.9 to –15.3 kcal/mol, which are comparable to or slightly better than the reference ligand Rifampin (–14.9 kcal/mol, shown in red). In contrast, the positive control ANP PC (blue) exhibits a considerably weaker binding affinity of –6.3 kcal/mol.

The molecular docking study performed against NfGlck demonstrated clear differences in binding affinities among the selected repurposed drugs and screened ligands. The docking analysis revealed that 64 novel ligands identified through virtual screening exhibited the highest binding affinities, with docking scores ranging from –16.8 to –15.0 kcal/mol, indicating strong and stable interactions within the ATP-binding pocket. Among the therapeutics, Rifampin showed the best binding affinity (–14.9 kcal/mol), suggesting a strong interaction with key active site residues and highlighting its potential as a lead compound. In addition, seven other ligands exhibited identical docking scores to Rifampin, reinforcing its relevance as a benchmark for comparison.

The reference ligand ANP displayed a docking score of –6.3 kcal/mol, confirming the validity and accuracy of the docking protocol. Other drugs such as Fluconazole (–6.2 kcal/mol), Azithromycin (–5.1 kcal/mol), and Miltefosine (–4.7

kcal/mol) showed comparatively weaker binding affinities, indicating less favorable interactions within the binding site.

5.9 Docking Protocol Validation

5.9.1 Validation of Re-Docked ANP and Co-Crystalized ANP ligand

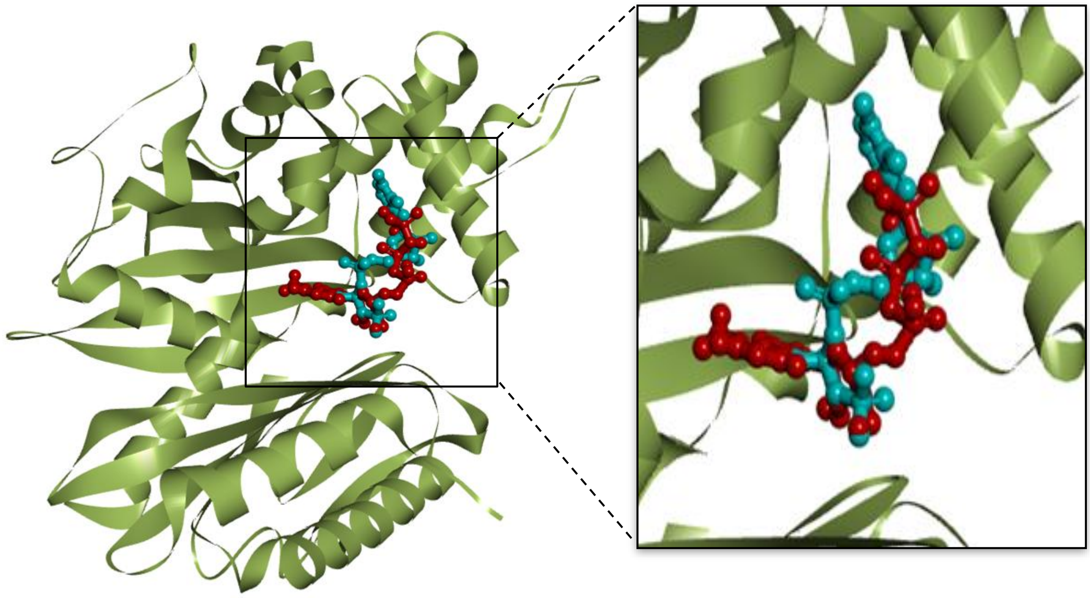


Figure 12: Superimposition of re-docked ANP (red) with the native co-crystallized ANP (blue) in the active site of NfGlck, indicating strong structural alignment.

Post docking analysis revealed overlap between the re-docked and native ANP conformations within the binding pocket of NfGlck (Figure 12). This high degree of superimposition suggests that the docking protocol reliably reproduces the experimentally observed binding mode. Such consistency validates the accuracy of the docking approach and supports its use for subsequent ligand screening and interaction analysis.

The validation of the docking protocol was performed by superimposing the re-docked ANP ligand (red) onto the co-crystallized native ANP ligand (blue) within the active site of NfGlck (Figure 12) using Discovery Studio Visualizer. The structural alignment demonstrates that the docked ligand closely overlaps with the native conformation, confirming that the docking procedure successfully reproduced the correct binding pocket and general ligand orientation. The grid box parameters (X = 76.70, Y = 44.62, Z = 7.78; dimensions: 7.87 Å × 11.42 Å × 12.52 Å) ensured that the docking search space was appropriately centered on the active site, allowing accurate placement of the ligand.

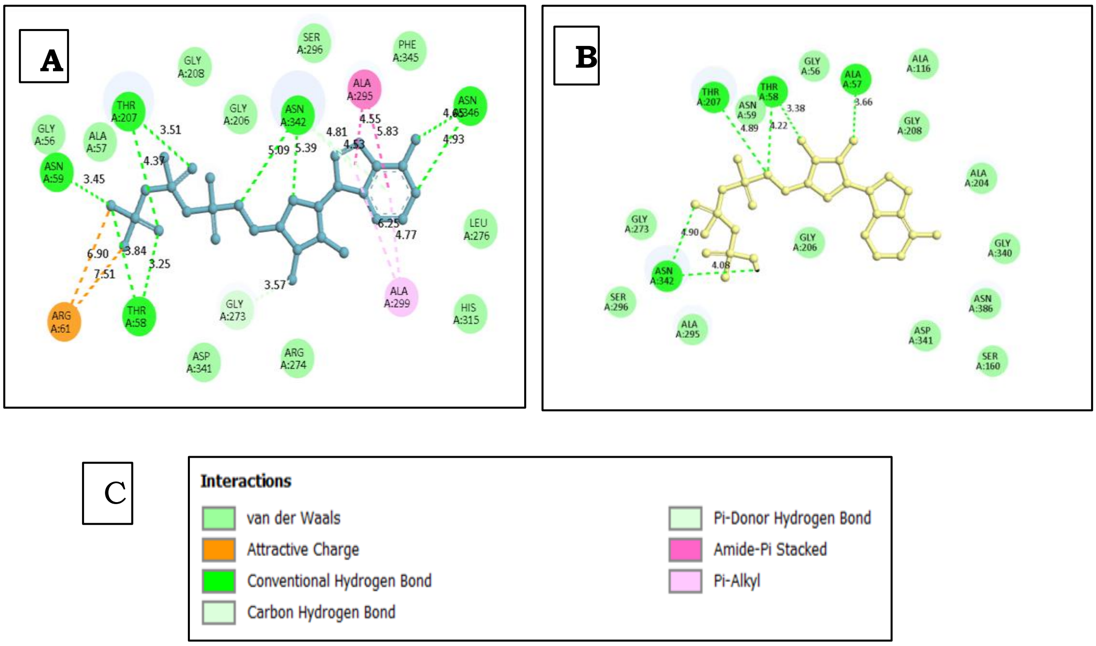


Figure 13: Comparative 2D interaction diagrams of native and re-docked ANP bound to NfGlck. (A) Native co-crystallized ANP (blue), (B) re-docked ANP (yellow), and (C) interaction

(Figure 13) (A) and (B) depict key binding interactions, including conventional hydrogen bonds (green lines) with bond lengths (Å), electrostatic interactions (orange lines), π-interactions (pink lines), and van der Waals contacts (light green regions). (C) presents the color-coded interaction types: van der Waals (light green), attractive charge (orange), conventional hydrogen bond (bright green), carbon hydrogen bond (pale green), π-donor hydrogen bond (grey-green), amide–π stacked (pink), and π-alkyl (light purple). The comparative 2D interaction analysis of native and re-docked ANP with NfGlck shows strong agreement in key binding interactions, supported by comparable bond lengths. Several critical residues were conserved across both complexes, indicating that the docking protocol effectively reproduced the core binding mode.

In the native structure, conventional hydrogen bonds were observed with ASN59 (3.45 Å), THR58 (3.25 Å), and THR207 (3.51 Å and 4.37 Å), along with additional interactions involving ASN342 (5.09 Å and 5.39 Å) and ASN346 (~4.63–4.93 Å). The re-docked complex retained similar interactions with THR58 (3.38 Å and 4.22 Å), THR207 (4.89 Å), ASN342 (4.08 Å and 4.90 Å), and ALA57 (3.66 Å). These distances (~3.2–5.4 Å) fall within an acceptable range, confirming preservation of key hydrogen-bonding geometry, particularly with the phosphate and ribose groups of ANP.

A notable difference is the presence of an electrostatic interaction between ARG61 and the phosphate group in the native complex (6.90 Å and 7.51 Å), which is absent in the docked structure, suggesting slight displacement of the phosphate moiety. Similarly, π-interactions (amide–π stacking and π-alkyl) involving ALA295 and ALA299 in the native complex (4.53–6.25 Å), stabilizing the adenine ring, were not retained after docking, indicating minor orientation changes.Van der Waals interactions involving GLY56, GLY206, GLY208, SER296, LEU276, PHE345, HIS315, and ASP341 were conserved in both complexes, contributing to structural stability through shape complementarity. The docking protocol reliably reproduced the primary binding interactions with comparable geometries. Minor deviations, including the loss of electrostatic and π-interactions, suggest slight positional differences, but the overall agreement supports the accuracy of the docking approach.

5.10 ADMET Analysis

The ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) analysis was performed to evaluate the pharmacokinetic properties and drug-likeness of the shortlisted ligands in comparison with the reference compound rifampin. Initially, a total of 64 ligands with binding affinities lower than –14.9 kcal/mol were selected for screening. Further filtering based on molecular weight (≤ 800 g/mol) resulted in the selection of seven ligands for detailed ADMET evaluation.

The reference compound rifampin exhibited a relatively high molecular weight (822.41 g/mol), whereas the selected ligands showed slightly lower molecular weights ranging from approximately 764.40 to 796.39 g/mol, indicating a moderate improvement in drug-likeness.

5.10.1 Physicochemical Properties

Physicochemical properties play a crucial role in determining the suitability of a compound as a drug candidate. According to Lipinski's Rule of Five, an ideal orally active drug should have a molecular weight ≤ 500 g/mol to ensure optimal permeability and absorption.

The reference rifampin violates this criterion due to its high molecular weight (822.41 g/mol). Similarly, all seven selected ligands also exceeded the recommended threshold, with molecular weights ranging from 764.40 to 796.39 g/mol. Despite this, such deviations are commonly observed in bulky antibiotic-like compounds. Both the reference compound and the selected ligands exhibited identical Lipinski's rule (score = 1), indicating that none of the compounds fully satisfied Lipinski's criteria.

The lipophilicity (logP) values of rifampin (~7.20) and the selected ligands (~7.10–7.25) were found to be comparable, suggesting similar membrane permeability characteristics. However, these high logP values also indicate increased hydrophobicity, which may negatively influence aqueous solubility. The topological polar surface area (TPSA) of rifampin (~220 Å²) was comparable to that of the selected ligands (190–234 Å²), reflecting similar polarity profiles. While this supports effective interaction with biological targets, elevated TPSA values may limit passive diffusion across biological membranes. Hydrogen bonding analysis revealed that rifampin (HBD ≈ 5; HBA ≈ 6) and the selected ligands (HBD: 4–8; HBA ≈ 6) possess comparable hydrogen bonding capabilities, indicating preserved interaction potential with the target protein.

5.10.2 Absorption and Distribution

Adequate absorption is essential for maintaining therapeutic efficacy. Ideally, compounds should exhibit a Human Intestinal Absorption (HIA) probability greater than 0.8 or an absorption percentage exceeding 30% for favorable oral bioavailability. Due to their large molecular size and high polarity, the predicted passive absorption values for all seven ligands were extremely low, closely resembling the absorption profile of rifampin. This suggests that, similar to the reference compound, the absorption of these ligands is likely mediated through active transport mechanisms rather than passive diffusion. The comparable lipophilicity values further indicate that membrane permeability characteristics are retained among the selected compounds and the reference drug.

5.10.3 Metabolism and Excretion

Metabolic stability and avoidance of adverse drug–drug interactions are critical factors in drug development. The results demonstrated that all selected ligands exhibited negligible inhibition of the CYP3A4 enzyme, with inhibition probabilities ranging from 0.001 to 0.016. These values are consistent with rifampin (0.009), indicating a favorable safety profile in terms of enzyme inhibition. However, both the reference drug (0.999) and the selected ligands (0.77–0.999) showed a high probability of being CYP3A4 substrates, suggesting that they may undergo similar metabolic pathways. In terms of excretion, the plasma clearance rates of the selected ligands (2.22–3.41) were slightly higher than that of rifampin (1.90), indicating relatively faster elimination. This may help reduce the risk of systemic accumulation and associated toxicity.

5.10.4 Toxicity Profile

Toxicity assessment is a critical step in determining the scientific viability of drug candidates. The selected ligands demonstrated excellent cardiac safety, with hERG inhibition probabilities ranging from 0.004 to 0.053, comparable to rifampin (0.006), thereby indicating a low risk of cardiotoxicity. The AMES mutagenicity and human hepatotoxicity predictions for the selected ligands showed profiles similar to that of the reference drug. Since hepatotoxicity is a known side effect of rifampin, the observed predictions for the selected compounds are consistent with the pharmacological characteristics of this drug class. Overall, the toxicity predictions indicated no significant additional toxicological risks for the selected ligands. Their safety profiles were found to be comparable to rifampin, and the high ADMET probability scores further support their acceptable pharmacokinetic behavior. Additional ADMET parameters are mentioned at Table 4, along with their ideal ranges in Table 5.

Table 4: Comparative ADMET profiling of selected ligand compounds, highlighting physicochemical properties, drug-likeness indices, pharmacokinetic behavior, and toxicity predictions for candidate evaluation.

PubChem ID	135985352	172943871	137117236	172967307	172950762	144663456	177831428	135398735-Rifampicin
Molecular Weight (g/mol)	764.40	778.38	780.39	780.39	780.39	790.42	796.39	822.41
Hydrogen Bond Acceptors	14.00	15.00	15.00	15.00	15.00	14.00	16.00	16.00
Hydrogen Bond Donors	7.0	6.0	7.0	7.0	7.0	5.0	8.0	6.0
Topological Polar Surface Area	204.85	210.92	214.08	214.08	214.08	190.69	234.31	220.15
Number of Rotatable Bonds	2.0	3.0	3.0	3.0	3.0	4.0	4.0	5.0
Number of Rings	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Number of Heteroatoms	14.0	15.0	15.0	15.0	15.0	14.0	16.0	16.0
Formal Charge	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
gasa (synthesizability)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
QED (Drug Likeness)	0.13	0.15	0.13	0.13	0.13	0.11	0.13	0.11
Synthetic Accessibility Score	7.14	7.14	7.15	7.15	7.15	7.14	7.25	7.20
Natural Product-likeness	1.22	1.51	1.41	1.41	1.41	1.52	1.39	1.66
Lipinski	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Pfizer	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
GSK	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Golden Triangle	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Solubility (logS)	-3.11	-3.07	-2.66	-2.96	-3.00	-3.43	-3.04	-3.33

Lipophilicity (logP)	1.59	1.51	1.12	1.50	1.48	2.45	1.26	1.49
pka_acidic	7.52	7.30	6.48	6.86	7.09	7.04	7.53	7.52
pka_basic	6.56	6.73	6.98	6.81	6.62	7.17	6.22	6.17
Caco2 Permeability	-5.41	-5.36	-5.60	-5.40	-5.40	-5.48	-5.49	-5.35
MDCK Permeability (Madin–Darby Canine Kidney cells)	-4.98	-4.97	-5.12	-5.04	-5.02	-4.98	-5.13	-4.94
PAMPA (Parallel Artificial Membrane Permeability)	1.0	1.0	1.0	1.0	1.0	0.9	1.0	1.0
P-glycoprotein (P-gp) Inhibitor	0.04	0.08	0.00	0.30	0.30	0.26	0.01	0.98
P-glycoprotein (P-gp) Substrate	1.0	0.92	1.0	1.0	1.0	0.96	1.0	1.0
Human Intestinal Absorption	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
f30 (Bioavailability at 30%)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Blood-Brain Barrier Penetration	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Plasma Protein Binding	85.43	83.92	85.59	84.04	84.10	87.04	81.32	84.07
Volume Distribution (logVDss)	0.46	0.47	0.55	0.50	0.51	0.55	0.37	0.44
CYP2C19-inhibitor	0.0	0.0	0.0	0.0	0.0	0.98	0.0	0.07
CYP2C19-substrate	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
CYP2C9-inhibitor	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CYP2C9-substrate	0.99	1.0	1.0	0.99	0.99	0.99	1.0	1.0
CYP2D6-inhibitor	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CYP2D6-substrate	0.0	0.01	0.92	0.0	0.0	0.01	0.0	0.09
CYP3A4-inhibitor	0.01	0.0	0.02	0.0	0.0	0.01	0.0	0.01
CYP3A4-substrate	1.0	1.0	1.0	1.0	1.0	0.77	1.0	1.0

hERG (Human Ether-à-go-go-Related Gene) Inhibitor	0.01	0.01	0.05	0.01	0.01	0.05	0.0	0.01
AMES (DNA Mutagenicity)	0.86	0.91	0.85	0.93	0.90	0.80	0.94	0.95
Liver Injury	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
FDA Maximum Recommended Daily Dose	0.54	0.78	0.96	0.40	0.45	0.99	0.55	0.60
Skin Sensitization	1.0	1.0	0.52	1.0	1.0	1.0	1.0	1.0
Carcinogenicity	0.97	0.98	0.99	0.99	0.98	0.80	0.98	0.98
Eye Corrosion	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Eye Irritation	0.26	0.10	0.0	0.23	0.33	0.15	0.18	0.13
Respiratory Toxicants	0.55	0.81	0.97	0.54	0.74	0.86	0.60	0.66
Human Hepatotoxicity	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Drug Induced Neurotoxicity	0.0	0.01	0.33	0.0	0.0	0.59	0.0	0.0
Ototoxicity	0.96	0.97	0.99	0.98	0.98	0.72	0.99	0.96
Hematotoxicity	0.58	0.62	0.44	0.75	0.58	0.31	0.62	0.60
Drug Induced Nephrotoxicity	0.98	0.99	0.85	0.98	0.98	0.99	0.99	0.98
Genotoxicity	1.0	1.0	0.95	1.0	1.0	1.0	1.0	1.0

Table 5: Optimal ranges for ADMET parameters obtained from ADMETlab analysis, encompassing physicochemical descriptors, medicinal chemistry properties, absorption, distribution, metabolism, excretion, and toxicity endpoints for assessing the drug-likeness.

Property	Optimal Range / Preferred Value
Molecular Weight (g/mol)	100–600
Hydrogen Bond Acceptors	0–12
Hydrogen Bond Donors	0–7
Topological Polar Surface Area (TPSA)	0–140 Å²
Number of Rotatable Bonds	0–11
Number of Rings	0–6
Number of Heteroatoms	1–15
Formal Charge	−4 to +4
gasa (synthesizeability)	Low / near 0
QED (Drug Likeness)	> 0.67 ideal; 0.49–0.67 acceptable
Synthetic Accessibility Score	< 6
Natural Product-likeness	−5 to +5
Lipinski	≤ 1 violation
Pfizer	0 (no violation)
GSK	1 (pass)
Golden Triangle	1 (pass)
Solubility (logS)	> −4
Lipophilicity (logP)	0–3
pKa (acidic)	3–11
pKa (basic)	6–10
Caco2 Permeability	> −5.15
MDCK Permeability	High > 20×10^{-6} cm/s
PAMPA	High (> 0.5)
P-gp Inhibitor	Low probability (≈ 0)
P-gp Substrate	Low–moderate
Human Intestinal Absorption	High (≥ 0.3)
f30 (Bioavailability 30%)	≥ 0.3
Blood-Brain Barrier Penetration	Context-dependent
Plasma Protein Binding (%)	< 90%
Volume Distribution (logVDss)	−0.15 to 1.5
CYP2C19 Inhibitor	Low (≈ 0)
CYP2C19 Substrate	Acceptable
CYP2C9 Inhibitor	Low (≈ 0)
CYP2C9 Substrate	Acceptable
CYP2D6 Inhibitor	Low (≈ 0)
CYP2D6 Substrate	Preferably low
CYP3A4 Inhibitor	Low (≈ 0)
CYP3A4 Substrate	Acceptable
hERG Blockers	Negative preferred
Liver injury	Negative preferred
AMES mutagenicity	Negative preferred
FDA Maximum Recommended Daily Dose	Within safe limits
Skin Sensitization	Low (≈ 0)
Carcinogenicity	Low (≈ 0)
Eye Corrosion	0
Eye Irritation	Low (≈ 0)
Respiratory Toxicants	Low (≈ 0)
Human Hepatotoxicity	Low (≈ 0)
Drug Induced Neurotoxicity	Low (≈ 0)
Ototoxicity	Low (≈ 0)

Hematotoxicity	Low (≈ 0)
Drug Induced Nephrotoxicity	Low (≈ 0)
Genotoxicity	Low (≈ 0)

5.11 Analysis of Top Candidates

5.11.1 Structural Convergence and Binding Surface of Selected Compounds

Interaction analysis of the selected lead candidates docked with NfGlck provides key insights into their binding stability and interaction patterns, supporting their potential as inhibitors. Seven top-performing compounds exhibited consistent and favorable binding within the ATP-binding pocket. Superimposition of the docked ligands showed strong spatial convergence, indicating a conserved binding mode (Figure 14). This clustering within the catalytic site highlights the compatibility of the binding pocket with diverse scaffolds and their ability to mimic essential ATP interactions.

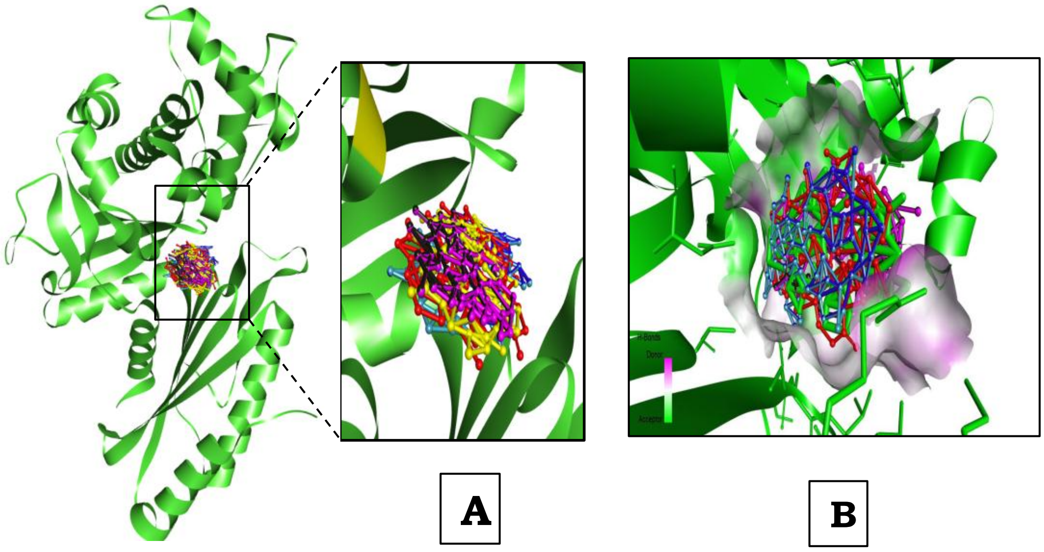


Figure 14: Docking visualization of ligand binding and hydrogen bonding.

(A) Superimposed poses of all docked ligands within the binding pocket, showing conformational overlap and spatial distribution. (B) Hydrogen bond interactions between ligands and protein residues, with donor (pink) and acceptor (light purple) regions

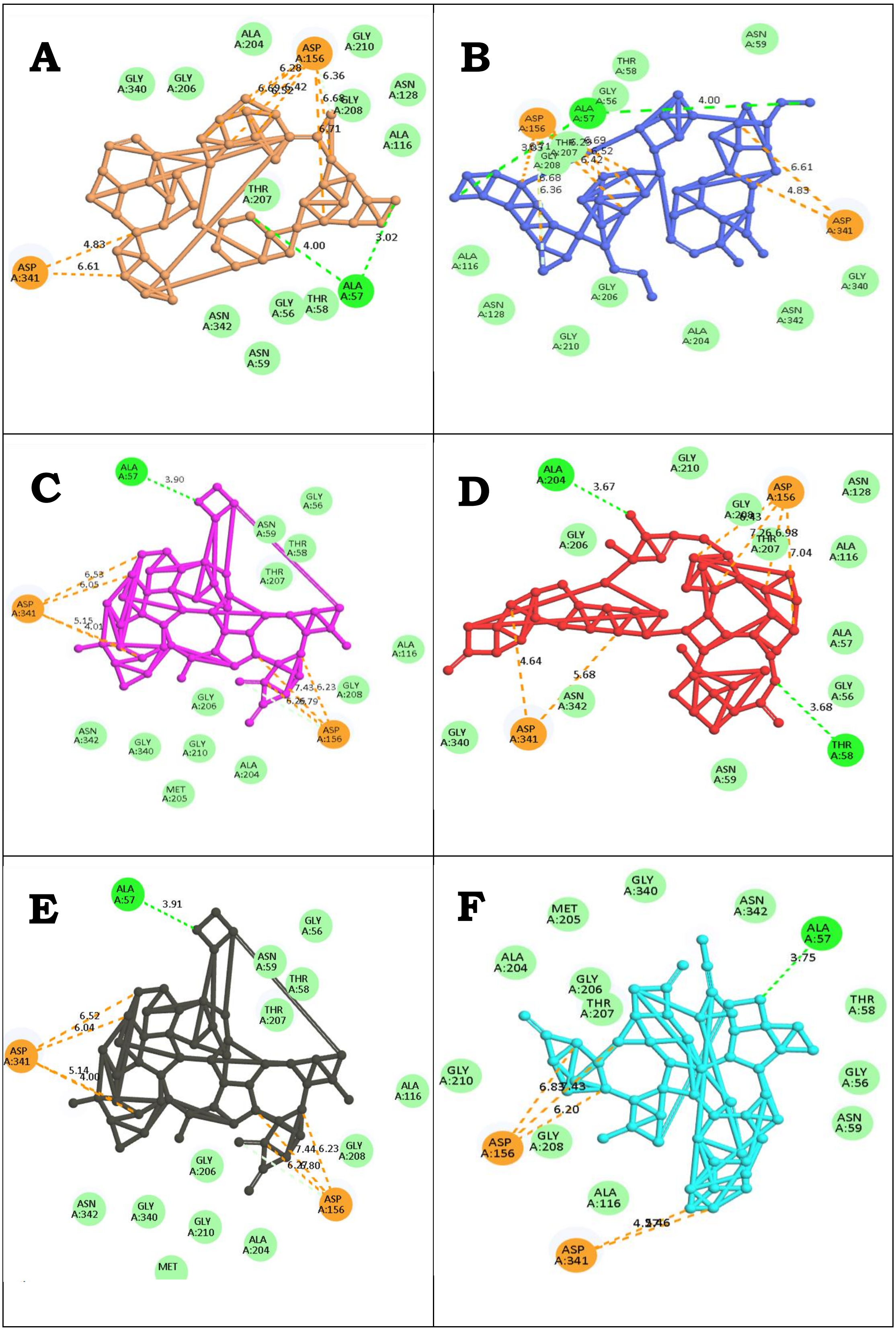
The binding pocket topography, visualized as a solvent-accessible surface, demonstrates strong spatial complementarity between the lead compounds and the protein. Color-coded mapping highlights hydrogen-bonding potential, with magenta indicating donor sites and green/light purple indicating acceptor sites. The clustered docking poses suggest a stable and energetically favorable binding mode, indicating effective active site occupancy and the potential for stronger intermolecular interactions compared to conventional treatments.

5.11.2 Comparative Interaction Profiling of Top Candidates vs. Native ANP

Structural analysis of the seven top-performing compounds (Figure 15) reveals distinct yet complementary interaction profiles. Compound B (Blue, ID: 135985352) forms a hydrogen bond with ALA57 (4.00 Å) and establishes strong attractive charge interactions with ASP156 (6.36–6.68 Å) and ASP341 (4.83–6.61 Å), while occupying a van der Waals pocket defined by GLY206, GLY210, and ALA204. Compound A (Orange, ID: 172943871) shows strong hinge-region anchoring through hydrogen bonds with ALA57 (3.02 Å and 4.00 Å) and enhanced electrostatic complementarity via multiple contacts with ASP156.

Compounds C (Pink, ID: 137117236) and E (Black, ID: 172967307) display nearly identical binding modes, anchored by ALA57 (~3.9 Å) and stabilized by a bifurcated electrostatic bridge involving ASP156 and ASP341 (4.00–7.44 Å). Compound F (Sky Blue, ID: 172950762) targets the phosphate-binding region, forming a hydrogen bond with ALA57 (3.75 Å) and charge interactions with ASP156 (6.20–6.83 Å) and ASP341 (4.27–4.46 Å), supported by van der Waals contacts with MET205 and ASN342. Compound D (Red, ID: 144663456) uniquely interacts with ALA204 (3.67 Å) and replicates the native THR58 hydrogen bond (3.68 Å), while engaging ASP156 and ASP341 for additional stabilization. Compound G (Yellow, ID: 177831428) maintains a hydrogen bond with ALA57 (3.52 Å) and a carbon hydrogen bond with ASN59 (5.01 Å), bridging the catalytic site through interactions with ASP156 (6.30–7.52 Å) and ASP341 (6.37 Å).

Across all candidates, consistent engagement of ASP156 and ASP341 represents a key finding. Although present in the native structure, these residues are not fully utilized by ANP for polar interactions. The selected ligands effectively exploit these sites, shifting stabilization from conventional hydrogen bonding toward stronger electrostatic interactions.


A
ALA A:204
ASP A:156
GLY A:210
GLY A:340
GLY A:206
ASN A:128
GLY A:208
ALA A:116
THR A:207
ASP A:341
ALA A:57
GLY A:56
THR A:58
ASN A:342
ASN A:59
B
ASN A:59
THR A:58
GLY A:56
ALA A:57
ASP A:156
THR A:207
GLY A:208
ASP A:341
ALA A:116
GLY A:206
GLY A:340
ASN A:128
ASN A:342
ALA A:204
GLY A:210
C
ALA A:57
GLY A:56
ASN A:59
THR A:58
THR A:207
ASP A:341
ALA A:116
GLY A:208
GLY A:206
ASP A:156
ASN A:342
GLY A:340
GLY A:210
ALA A:204
MET A:205
D
ALA A:204
GLY A:210
ASP A:156
ASN A:128
GLY A:206
THR A:207
ALA A:116
ALA A:57
GLY A:56
ASN A:342
ASP A:341
GLY A:340
THR A:58
ASN A:59
E
ALA A:57
GLY A:56
ASN A:59
THR A:58
THR A:207
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ALA A:116
GLY A:208
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ASN A:342
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ALA A:204
MET
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MET A:205
ASN A:342
ALA A:57
ALA A:204
GLY A:206
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ALA A:116
ASP A:341

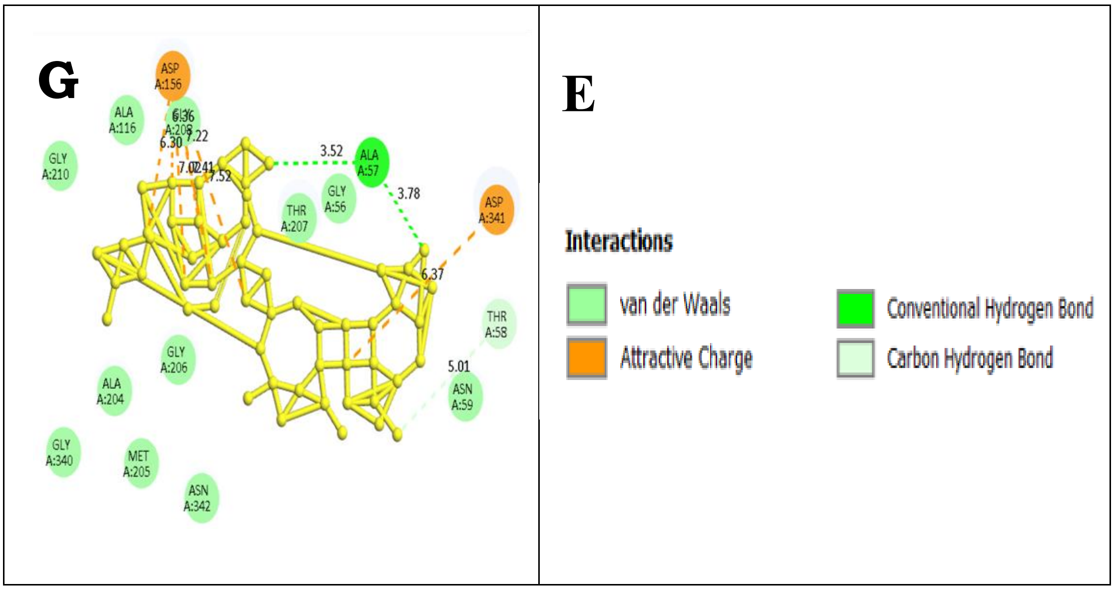
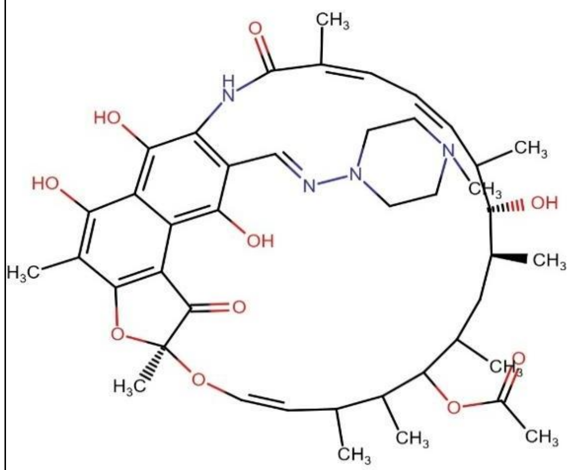
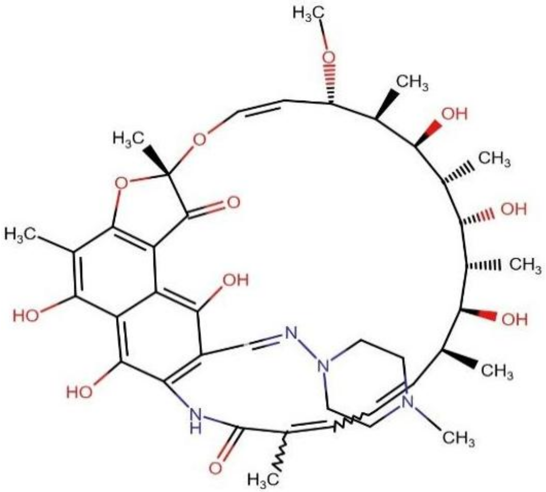


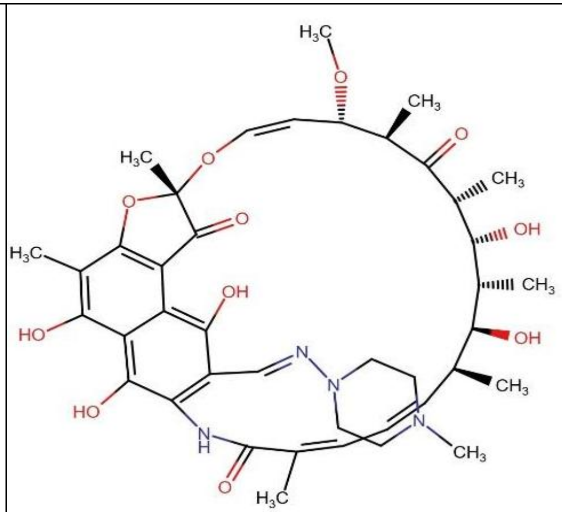
Figure 15: Comprehensive 2D interaction analysis of selected compounds (A–G) within the active site of NfGlck.

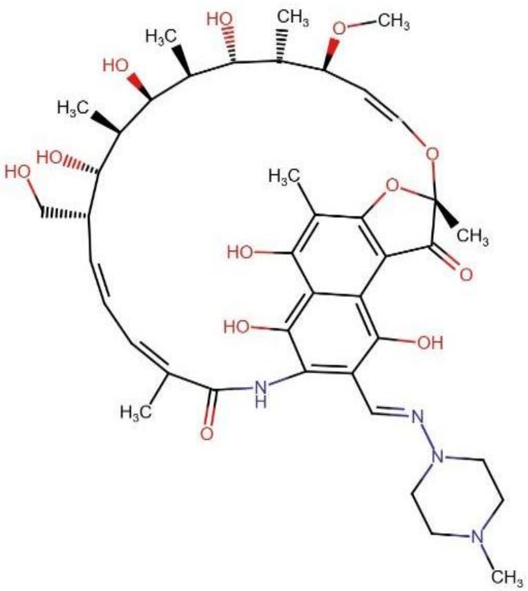
This interaction pattern, combined with conserved van der Waals support from residues such as GLY206, GLY208, and GLY210, suggests enhanced binding affinity and potential residence time. Collectively, these results identify the seven compounds as promising candidates for further in vitro evaluation against *Naegleria fowleri* glucokinase. Table 6, shows ligand’s structure, binding affinity, interaction along with distance, bond type and van der Waal’s interactions.

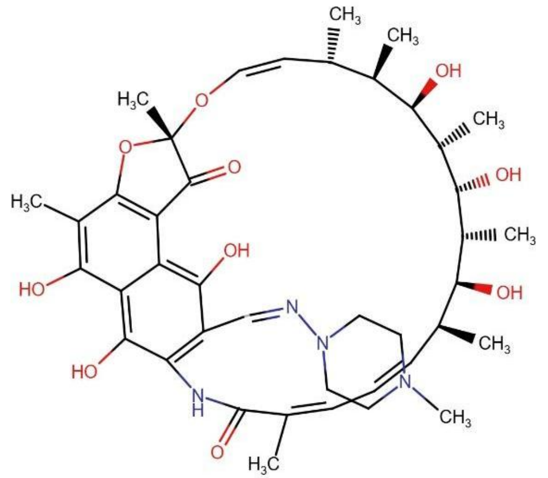
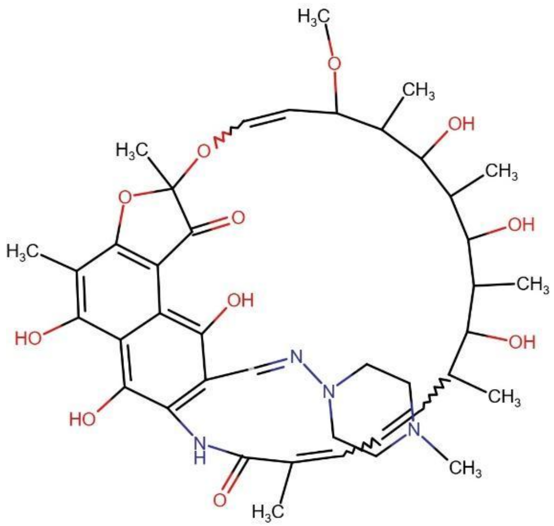
Table 6: *The binding affinities, compound structures, interactions, distances, van der Waals interaction details of selected compounds Post-docking and ADMET*

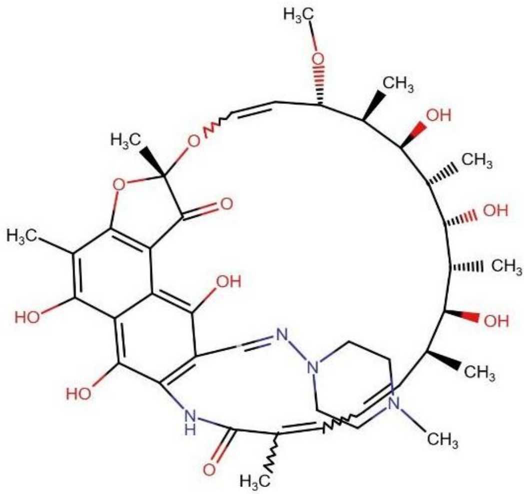
Compound ID	**Structure**	**Binding affinity (kcal/mol)**	**Interactions**	**Distance (Å)**	**Bond Type**	**Van der Waal's Interactions (VdW)**
144663456		-15.0	144663456:O - ASP341:OD2	5.36	Electrostatic	Gly56, Ala57, Asn59, Ala116, Asn128, Gly206, Thr207, Gly208, Gly210, Gly340, and Asn342.
			144663456:O - ASP156:OD2	4.54	Electrostatic	
			144663456:O - ASP156:OD2	3.75	Electrostatic	
			144663456:O - ASP156:OD2	4.57	Electrostatic	
			144663456:O - ASP156:OD2	4.83	Electrostatic	
			144663456:O - ASP341:OD2	4.52	Electrostatic	
			144663456:OàALA204:O	2.77	Hydrogen	
			THR58:HNà144663456:O	2.11	Hydrogen	
172950762		-14.9	172950762:O ASP341:OD2	4.10	Electrostatic	Gly56, Thr58, Asn59, Ala116, Ala204, Met205,
			172950762:O ASP156:OD2	4.37	Electrostatic	
			172950762:O - ASP156:OD2	3.51	Electrostatic	

			172950762:O - ASP156:OD2	5.00	Electrostatic	Gly206, Thr207, Gly208, Gly210, Gly340, and Asn342.
			172950762: N - ASP341:OD2	4.61	Electrostatic	
			ALA57:HN à172950762:O	2.12	Hydrogen	
172943871		-15.3	172943871:O - ASP156:OD2	4.16	Electrostatic	Gly56, Thr58, Asn59, Ala116, Asn128, Ala204, Gly206, Gly208, Gly210, Gly340, and Asn342.
			172943871:O - ASP341:OD2	3.9	Electrostatic	
			172943871:O - ASP341:OD2	5.39	Electrostatic	
			172943871:O - ASP156:OD2	3.61	Electrostatic	
			172943871:O - ASP156:OD2	3.75	Electrostatic	
			172943871: N - ASP156:OD2	4.19	Electrostatic	
			172943871: N - ASP156:OD2	4.23	Electrostatic	

			172943871: N - ASP156:OD2	3.92	Electrostatic	
			ALA57:HN à172943871:O	1.68	Hydrogen	
			ALA57:HN à 172943871:O	2.42	Hydrogen	
			THR207:HG1 à172943871:O	3.08	Hydrogen	
			172943871:Cà ASP156:OD2	3.77	Hydrogen	
177831428		-14.9	177831428:O - ASP156:OD2	4.67	Electrostatic	Gly56, Thr58, Asn59, Ala116, Ala204, Met205, Gly206, Gly208, Gly210, Asn342, and Gly340
			177831428:O - ASP156:OD2	3.68	Electrostatic	
			177831428:O - ASP341:OD2	5.12	Electrostatic	
			177831428:O - ASP156:OD2	4.73	Electrostatic	
			177831428: N - ASP156:OD2	4.46	Electrostatic	
			177831428: N - ASP156:OD2	3.68	Electrostatic	
			177831428: N - ASP156:OD2	5.11	Electrostatic	

			ALA57:HN à177831428:O	1.98	Hydrogen	
			ALA57:HN à177831428:O	2.32	Hydrogen	
			THR207:HG1 à 177831428:O	2.94	Hydrogen	
135985352		-15	135985352:O - ASP156:OD2	4.15	Electrostatic	Gly56, Thr58, Asn59, Ala116, Asn128, Ala204, Gly206, Thr207, Gly208, Gly210,
			135985352:O - ASP341:OD2	5.39	Electrostatic	
			135985352:O - ASP341:OD2	3.9	Electrostatic	
			135985352:O - ASP156:OD2	3.61	Electrostatic	
			135985352: N - ASP156:OD2	3.75	Electrostatic	

			135985352: N - ASP156:OD2	4.19	Electrostatic	Gly340, and Asn342.
			135985352: N - ASP156:OD2	4.23	Electrostatic	
			135985352: N - ASP156:OD2	3.92	Electrostatic	
			ALA57:HN à135985352:O	1.69	Hydrogen	
			135985352:C àASP156:OD2	3.77	Hydrogen	
137117236		-15	137117236:O - ASP341:OD2	3.98	Electrostatic	Gly56, Thr58, Asn59, Ala116, Asn128, Ala204, Met205, Gly206, Thr207, Gly208, Gly210, Gly340, and Asn342.
			137117236:O - ASP341:OD2	5.59	Electrostatic	
			137117236:O - ASP156:OD2	4.32	Electrostatic	
			137117236:O - ASP156:OD2	3.53	Electrostatic	
			137117236:O - ASP156:OD2	5.01	Electrostatic	
			137117236: N - ASP341:OD2	5.54	Electrostatic	
			137117236: N - ASP341:OD2	4.39	Electrostatic	
			ALA57:HN à137117236:O	2.33	Hydrogen	

			137117236:C àASP156:OD2	3.69	Hydrogen	
172967307		-15	172967307:O - ASP341:OD2	3.98	Electrostatic	Gly56, Thr58, Asn59, Ala204, Met205, Gly206, Thr207, Gly208, Gly210, Gly340, and Asn342.
			172967307:O - ASP341:OD2	5.58	Electrostatic	
			172967307:O - ASP156:OD2	4.33	Electrostatic	
			172967307:O - ASP156:OD2	3.53	Electrostatic	
			172967307:O - ASP156:OD2	5.02	Electrostatic	
			172967307: N - ASP341:OD2	5.53	Electrostatic	
			172967307: N - ASP341:OD2	4.38	Electrostatic	
			ALA57:HN à172967307:O	2.34	Hydrogen	
			172967307:C àASP156:OD2	3.7	Hydrogen	

5.12 Validation of stability analysis

The RMSF graphs were generated from simulations performed using CABS-flex for the top compounds selected after molecular docking with the NfGlck protein. This analysis enabled evaluation of residue-level flexibility and binding-site stability in comparison with the native ligand and the reference drug Rifampin. Overall, this approach was used to assess the dynamic stability of the selected lead compounds and their influence on protein flexibility during simulation.

5.12.1 Validation of Docked ANP Dynamic Stability with Native ANP

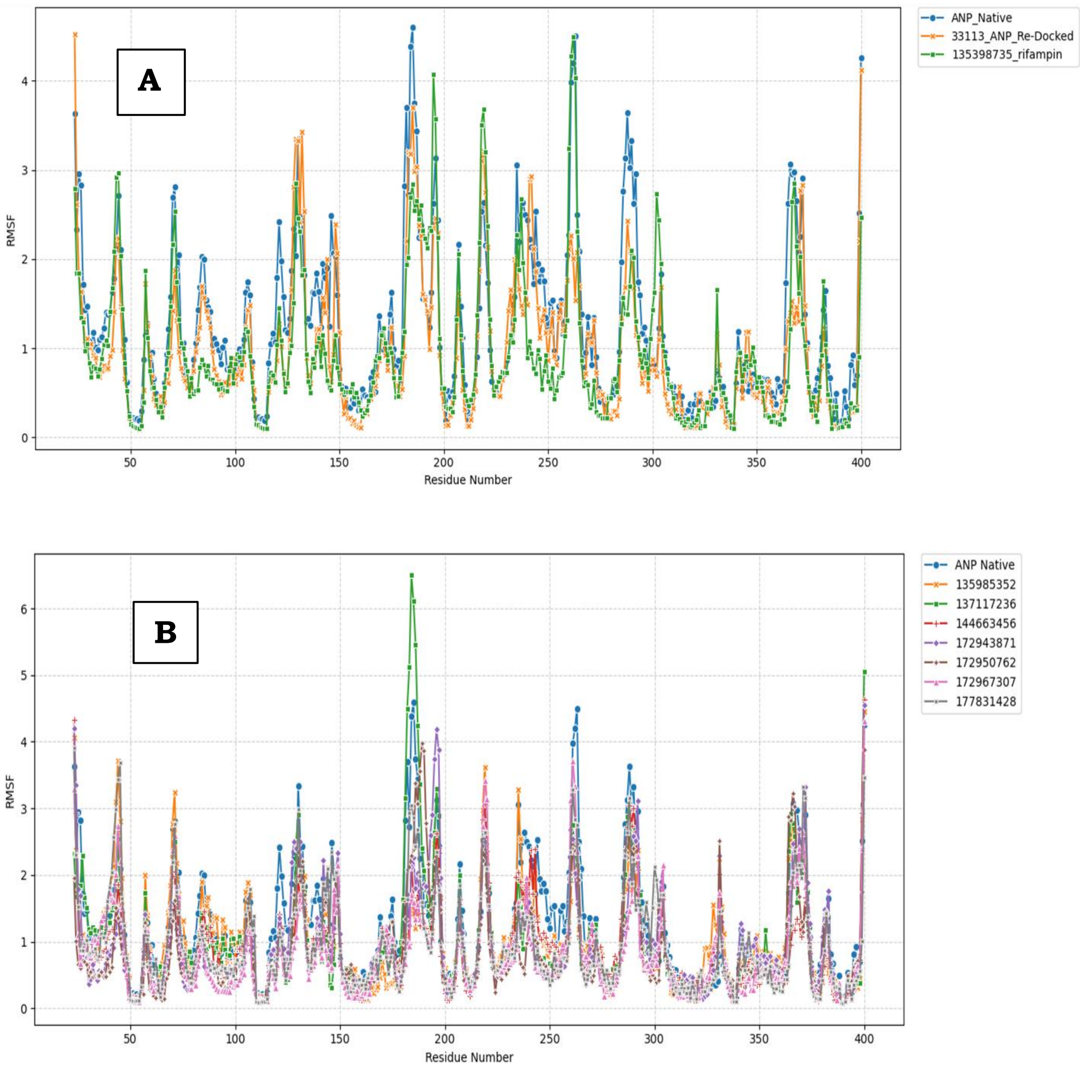


Figure 16: MDS done for 500 ns (A) RMSF profile showing residue-level fluctuations of NfGlck bound to native ANP (blue), re-docked ANP (orange), and the reference drug Rifampin. (green). (B) RMSF profile of NfGlck complexes with selected screened compounds (135985352, 137117236, 144663456, 172943871, 172950762, 172967307, and 177831428) compared with the native ANP system (blue).

As an initial step, this approach was used to validate the reliability of the docking protocol. The RMSF analysis demonstrated that the native co-crystallized

(33113_ANP) ligand and re-docked ANP ligand (orange) (Figure 16) closely reproduce the dynamic simulations of the native ANP system (blue). The fluctuation patterns of both systems showed strong overlap across the majority of residues throughout the protein structure. Both the native ANP complex (blue) and the re-docked ANP complex (orange) exhibited similar RMSF trends along the protein backbone, indicating that the ligand maintained a binding orientation consistent with the crystallographic conformation. Regions displaying higher flexibility were observed around residues 170–210 and 240–270, suggesting that these regions represent intrinsic mobility hotspots within the protein structure rather than ligand-induced instability.

Importantly, the re-docked complex (orange) did not exhibit abnormal fluctuations or large deviations from the native system (blue), indicating that the docking procedure preserved the native interaction network within the binding pocket. The close agreement between the RMSF profiles of the native ANP system (blue) and the re-docked ANP complex (orange) therefore confirms that the docking protocol can reliably reproduce biologically relevant ligand binding conformations and is suitable for further virtual screening and lead identification studies.

5.12.2 Comparative Dynamic Analysis of the Native ANP with selected Compounds

Following validation of the docking protocol, RMSF analysis was extended to evaluate the dynamic behavior of the NfGlck protein in complex with the screened compounds. The RMSF profiles of the selected ligands were compared with the native ANP system (blue) to assess their ability to stabilize the protein structure during the 500 ns molecular dynamics simulation. The majority of the screened compounds exhibited RMSF patterns broadly similar to the native ANP complex (blue), indicating that the overall structural stability of the protein was maintained throughout the simulation. However, variations in residue fluctuations were observed among the complexes, reflecting differences in the strength and nature of protein–ligand interactions within the binding pocket.

The reference drug Rifampin (green) displayed moderate residue fluctuations across several regions of the protein, particularly around the flexible segments identified near residues 170–210 and 240–270. While the Rifampin complex (green) maintained global stability of the protein, certain screened compounds demonstrated comparatively reduced fluctuations in these regions, suggesting stronger stabilization of the protein structure. Notably, compounds 137117236 (green dashed), 144663456 (red), and 172967307 (pink) showed RMSF profiles comparable to or slightly lower than the native ANP system (blue) across several regions of the protein backbone. The reduced fluctuations observed for these complexes indicate enhanced rigidity of the protein–ligand system and suggest favorable interactions within the binding pocket.

Overall, the comparative RMSF analysis indicates that several screened compounds maintain dynamic stability comparable to the native ligand while providing improved stabilization in specific regions of the protein.

5.12.3 Binding Site-Level RMSF Analysis of Key Residues

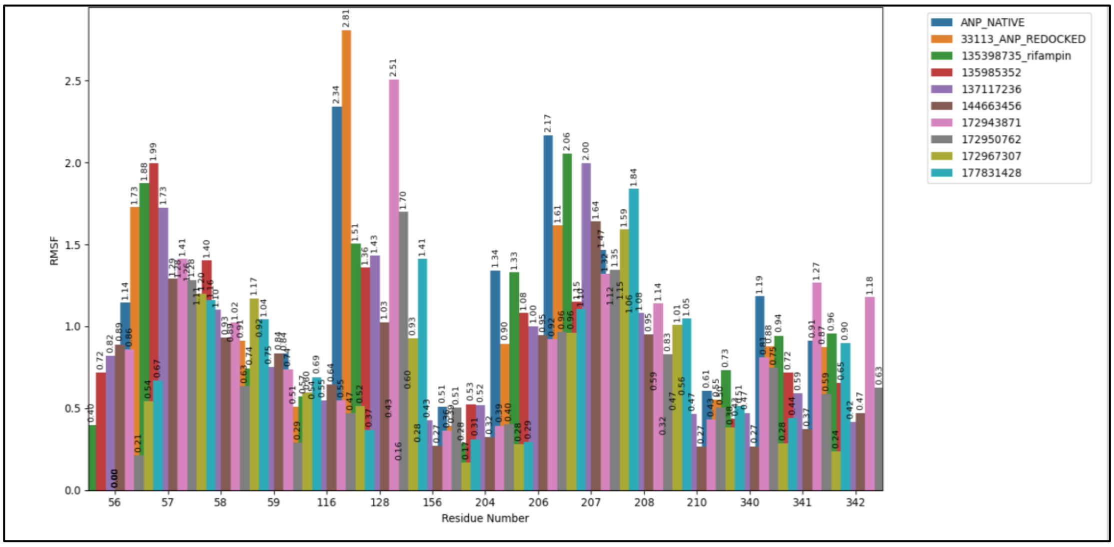


Figure 17: Binding site-level RMSF analysis of key residues in NfGlck–ligand complexes.

Bar plot (Figure 17) showing RMSF values for important binding-site residues (GLY56, ALA57, THR58, ASN59, ALA116, ASN128, ASP156, ALA204, GLY206, THR207, GLY208, GLY210, GLY340, ASP341, and ASN342) of NfGlck in complexes with native ANP, re-docked ANP, the reference drug Rifampin, and screened compounds.

To further investigate ligand-induced stability at the catalytic pocket of NfGlck, RMSF values were examined for key binding-site residues including GLY56, ALA57, THR58, ASN59, ALA116, ASN128, ASP156, ALA204, GLY206, THR207, GLY208, GLY210, GLY340, ASP341, and ASN342. These residues are located within or in close proximity to the ligand binding region and therefore play an important role in stabilizing protein–ligand interactions.

The native ANP complex (blue) exhibited moderate fluctuations across these residues, reflecting the dynamic nature of the binding pocket. The re-docked ANP complex (orange) showed RMSF values highly comparable to the native system, further confirming the reliability of the docking protocol. Comparison with the screened compounds revealed that several ligands were able to maintain or reduce fluctuations at these key residues. In particular, compounds 137117236 (green dashed), 144663456 (red), and 172967307 (pink) demonstrated relatively lower RMSF values across multiple binding-site residues compared with the reference drug Rifampin (green). The reduced fluctuations indicate stronger stabilization of the local protein environment and suggest favorable interactions between the ligands and binding pocket residues.

Among the analyzed compounds, 137117236 (green dashed) showed the most consistent stabilization across multiple residues within the binding pocket, while 144663456 (red) and 172967307 (pink) also maintained relatively low fluctuations in several key regions. These observations suggest that the identified compounds can effectively stabilize the structural architecture of the NfGlck active site during molecular dynamics simulation. Overall, the binding-site RMSF analysis supports the selection of 137117236 (green dashed), 144663456 (red), and 172967307 (pink) as promising lead candidates due to their ability to maintain reduced residue mobility and enhanced structural stability within the catalytic pocket.

5.13 DFT-Based FMO Analysis

The DFT-based FMO analysis revealed distinct variations in the electronic properties of the proposed ligands compared to the reference compound Rifampin (135398735). The HOMO–LUMO energy gap (ΔE), along with global reactivity descriptors such as hardness (η), softness (S), electronegativity (χ), and electrophilicity index (ω), were used to evaluate the electronic behavior of the compounds (Table 7). Among the studied molecules, ligand 172967307 exhibited the lowest energy gap (ΔE = -2.1061) and highest electrophilicity (ω = 0.1806 eV), indicating enhanced reactivity and electron-accepting potential. Ligand 172950762 also showed a relatively low ΔE with favorable softness, suggesting balanced reactivity and stability. Ligand 137117236 demonstrated moderate ΔE, hardness, softness, electronegativity, and electrophilicity values, supporting stable interaction potential. In contrast, rifampin (135398735) displayed a comparatively higher ΔE along with moderate hardness, softness, electronegativity, and electrophilicity, indicating lower electronic reactivity.

Ligands 172967307, 172950762, and 137117236 showed improved electronic profiles—characterized by lower ΔE and favorable hardness, softness, electronegativity, and electrophilicity—highlighting their potential as promising candidates for further investigation.

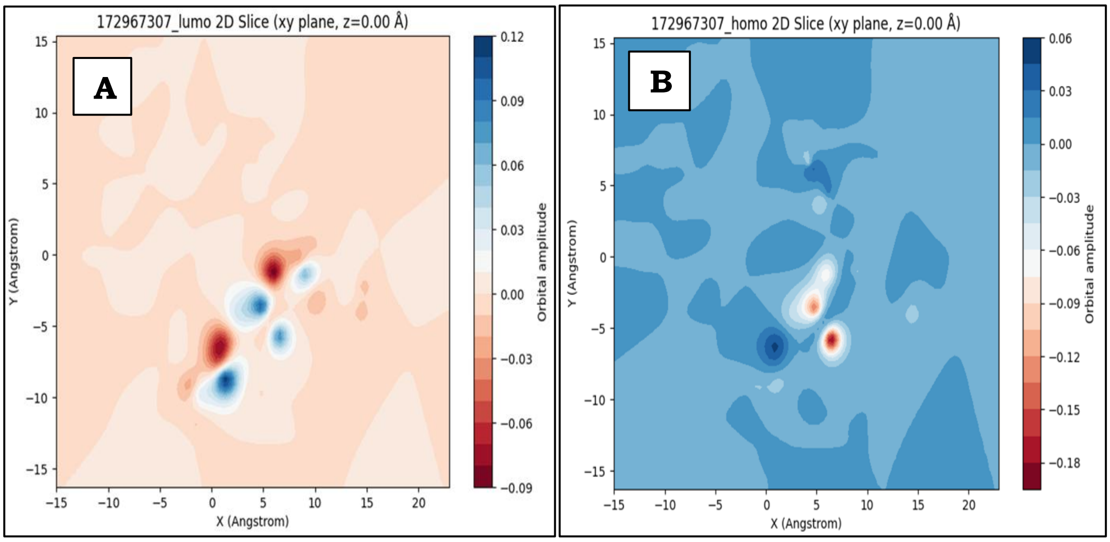


Figure 18 (A) HOMO orbital 2D slice (xy plane, z = 0.00 Å) showing electron-donating regions of the molecule. **(B)** LUMO orbital 2D slice (xy plane, z = 0.00 Å) indicating electron-accepting regions and potential sites for charge transfer.

Table 7: Frontier molecular orbital (FMO) descriptors and global reactivity parameters of selected compounds.

Pubchem ID	**Homo (eV)**	**Lumo (eV)**	**Gap (ΔE) (eV)**	**Hardness (η) eV**	**Softness (S) eV^-1**	**Electro negativity (χ) eV**	**Electrophilicity (ω) eV**
135985352	-2.4156	0.9981	3.4137	1.7069	0.5859	0.7087	0.1471
172943871	-1.9092	1.0104	2.9196	1.4598	0.685	0.4494	0.0692
137117236	-2.0618	0.8858	2.9476	1.4738	0.6785	0.588	0.1173
172967307	-2.1061	0.6858	2.7919	1.3959	0.7164	0.7101	0.1806
172950762	-2.0358	0.8517	2.8875	1.4437	0.6926	0.592	0.1214
144663456	-1.9901	0.9321	2.9222	1.4611	0.6844	0.529	0.0958
177831428	-1.9491	0.9401	2.8892	1.4446	0.6922	0.5045	0.0881
33113-ANP	-2.5503	1.1664	3.7167	1.8583	0.5381	0.6919	0.1288
135398735-rifampin	-2.4697	0.9389	3.4086	1.7043	0.5867	0.7654	0.1719

CHAPTER- 6

CONCLUSION

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CONCLUSION

This study implemented an integrated *in-silico* drug discovery pipeline to identify potential inhibitors targeting glucokinase (NfGlck), a key enzyme essential for glycolysis and survival of *N. fowleri*. Glucokinase catalyzes the ATP-dependent phosphorylation of glucose to glucose-6-phosphate, representing the first committed step of glycolysis and a critical regulatory point in the parasite's central carbon metabolism. Owing to this indispensable functional role, NfGlck was selected as a strategic therapeutic target. A set of repurposed compounds—Amphotericin B (5280965), Azithromycin (447043), Rifampin (135398735), Miltefosine (3599), and Fluconazole (3365)—was selected based on prior evidence (Dai et al., 2025) and evaluated alongside the native co-crystallized ligand ANP (33113), representing the ATP-bound state of the enzyme. A total of 2,064 compounds were retrieved from the PubChem analog library, of which 2,018 were successfully prepared through SDF→PDBQT conversion and UFF-based energy minimization using PyRx V0.8. Among the reference compounds, Rifampin exhibited the strongest binding affinity (–14.9 kcal/mol), serving as a benchmark for subsequent analyses. Virtual screening identified 64 compounds with superior docking scores (–16.8 to –15.0 kcal/mol), indicating strong and stable binding within the ATP-binding pocket.

ADMET profiling was also performed to evaluate the pharmacokinetic properties of the screened candidates and support their drug-likeness assessment. Subsequent filtering based on physicochemical criteria, including a molecular weight threshold (≤800 g/mol), refined the dataset to seven promising candidates. Residue-level interaction analysis revealed consistent involvement of key conserved residues, particularly GLY56, GLY206, and GLY208, across multiple ligand–protein complexes, highlighting their critical role in stabilizing ligand binding within the active site. These candidates established stable hydrogen bonding and hydrophobic interactions, contributing to binding specificity and structural integrity. CABS-flex RMSF analysis further demonstrated reduced residue-level fluctuations compared to ANP and Rifampin, indicating enhanced conformational stability of the protein–ligand complexes. Quantum chemical analysis revealed that ligands 172967307, 172950762, and 137117236 possess lower HOMO–LUMO energy gaps, improved hardness–softness balance, and higher electrophilicity relative to Rifampin, suggesting enhanced electronic reactivity and interaction potential. Collectively, the convergence of strong binding affinities (–16.8 kcal/mol), consistent interactions with conserved active-site residues, reduced RMSF deviations, and optimized electronic descriptors highlights these compounds as promising NfGlck inhibitors. This study establishes a robust computational framework that provides a strong foundation for experimental validation and advances the development of targeted therapeutics against N. *fowleri* infections.

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APPENDIX

APPENDIX: Python Script for DFT-Based FMO Analysis

The following Python script was used to perform density functional theory (DFT)-based analysis of selected compounds, including HOMO–LUMO energy calculations, molecular electrostatic potential (MEP) mapping, dipo, and global reactivity descriptor evaluation. The workflow integrates RDKit for molecular structure generation and optimization, PySCF for quantum chemical computations, and py3Dmol for three-dimensional visualization.

```
!pip install rdkit pyscf py3Dmol
import numpy as np
import matplotlib.pyplot as plt
import py3Dmol
import time
import os
from rdkit import Chem
from rdkit.Chem import AllChem
from rdkit.Chem.rdmolops import GetFormalCharge
from pyscf import gto
from pyscf.tools import cubegen

# Attempt to import GPU support; fall back to CPU if not available
try:
    from gpu4pyscf import dft as gpu_dft
    dft = gpu_dft
    print("Using GPU for DFT via gpu4pyscf")
except ImportError:
    from pyscf import dft
    print("gpu4pyscf not installed; using CPU for DFT. Install with: !pip install
gpu4pyscf")

def compute_mep(pyscf_mol, mf, label):
    try:
        os.makedirs('/kaggle/working/orbitals', exist_ok=True)
        mep_file = f'/kaggle/working/orbitals/{label}_mep.cube'
        cubegen.density(pyscf_mol, mep_file, mf.make_rdm1())
        print(f"MEP cube file saved: {mep_file}")
        return mep_file
    except Exception as e:
        print(f"Error generating MEP for {label}: {e}")
        return None

def compute_dipole(pyscf_mol, mf, label):
    try:
        dipole = mf.dip_moment(unit='Debye')
        print(f"{label} Dipole Moment (Debye): {dipole}")
        return dipole
    except Exception as e:
        print(f"Error computing dipole for {label}: {e}")
        return None

def compute_reactivity_descriptors(homo_ev, lumo_ev, label):
    try:
        hardness = (lumo_ev - homo_ev) / 2
        softness = 1 / hardness if hardness != 0 else float('inf')
        electronegativity = -(homo_ev + lumo_ev) / 2
        electrophilicity = (electronegativity ** 2) / (2 * hardness) if hardness !=
0 else float('inf')
        print(f"{label} Reactivity Descriptors:")
        print(f"Chemical Hardness (η): {hardness:.4f} eV")
        print(f"Chemical Softness (1/η): {softness:.4f} eV^-1")
        print(f"Electronegativity (χ): {electronegativity:.4f} eV")
```

```
        print(f"Electrophilicity (ω): {electrophilicity:.4f} eV")
        return hardness, softness, electronegativity, electrophilicity
    except Exception as e:
        print(f"Error computing reactivity descriptors for {label}: {e}")
        return None, None, None, None

def save_summary(label, homo_ev, lumo_ev, gap_ev, dipole, hardness, softness,
electronegativity, electrophilicity):
    try:
        os.makedirs('/kaggle/working/orbitals', exist_ok=True)
        with open(f'/kaggle/working/orbitals/{label}_summary.txt', 'w') as f:
            f.write(f"{label} Computational Summary\n")
            f.write(f"HOMO: {homo_ev:.4f} eV\n")
            f.write(f"LUMO: {lumo_ev:.4f} eV\n")
            f.write(f"Gap: {gap_ev:.4f} eV\n")
            if dipole is not None:
                f.write(f"Dipole Moment: {dipole}\n")
            if hardness is not None:
                f.write(f"Chemical Hardness: {hardness:.4f} eV\n")
                f.write(f"Chemical Softness: {softness:.4f} eV^-1\n")
                f.write(f"Electronegativity: {electronegativity:.4f} eV\n")
                f.write(f"Electrophilicity: {electrophilicity:.4f} eV\n")
        print(f"Summary saved to /kaggle/working/orbitals/{label}_summary.txt")
    except Exception as e:
        print(f"Error saving summary for {label}: {e}")

def compute_homo_lumo_and_save(smiles, label, basis='sto-3g'):
    start_time = time.time()

    # Parse SMILES and compute charge
    mol = Chem.MolFromSmiles(smiles)
    if mol is None:
        print(f"Invalid SMILES for {label}")
        return False
    charge = GetFormalCharge(mol)
    num_atoms = mol.GetNumAtoms()
    print(f"Processing {label}: {num_atoms} atoms, charge {charge}")

    # Generate 3D conformation with robust embedding
    mol = Chem.AddHs(mol)
    embed_params = AllChem.ETKDGv3()
    embed_params.maxIterations = 1000
    embed_params.randomSeed = 42
    embed_params.useRandomCoords = True
    embed_status = AllChem.EmbedMolecule(mol, embed_params)

    if embed_status != 0:
        print(f"Initial embedding failed for {label}. Trying UFF optimization...")
        try:
            AllChem.UFFOptimizeMolecule(mol, maxIters=1000)
            if mol.GetConformer().Is3D():
                print("UFF optimization successful.")
            else:
                print(f"UFF optimization failed to produce 3D structure for
{label}.")
                return False
        except Exception as e:
            print(f"UFF optimization failed for {label}: {e}")
            print("Consider using external tools like OpenBabel or CREST.")
            return False

    # Optimize with MMFF
    try:
        AllChem.MMFFOptimizeMolecule(mol, maxIters=1000)
        print("MMFF optimization successful.")
    except Exception as e:
```

```
        print(f"MMFF optimization failed for {label}: {e}, proceeding with current
geometry.")

    # Extract XYZ coordinates for DFT
    conf = mol.GetConformer()
    xyz = ''
    for atom in mol.GetAtoms():
        pos = conf.GetAtomPosition(atom.GetIdx())
        xyz += f"{atom.GetSymbol()} {pos.x:.4f} {pos.y:.4f} {pos.z:.4f}\n"

    # Build PySCF molecule
    pyscf_mol = gto.M(atom=xyz, basis=basis, charge=charge, spin=0)

    # Run DFT (B3LYP)
    mf = dft.RKS(pyscf_mol)
    mf.xc = 'b3lyp'
    mf.conv_tol = 1e-8
    mf.max_cycle = 200
    try:
        energy = mf.kernel()
        if not mf.converged:
            print(f"DFT calculation did not converge for {label}. Trying PBE
functional...")
            mf.xc = 'pbe'
            energy = mf.kernel()
            if not mf.converged:
                print(f"DFT calculation failed for {label} with PBE.")
                return False
    except Exception as e:
        print(f"DFT calculation failed for {label}: {e}")
        return False

    # Compute dipole moment
    dipole = compute_dipole(pyscf_mol, mf, label)

    # Generate MEP
    mep_file = compute_mep(pyscf_mol, mf, label)

    # Get HOMO and LUMO indices
    mo_energy = mf.mo_energy
    mo_occ = mf.mo_occ
    homo_idx = None
    lumo_idx = None
    for i in range(len(mo_occ)):
        if mo_occ[i] > 0:
            homo_idx = i
        elif mo_occ[i] == 0 and lumo_idx is None:
            lumo_idx = i
    if homo_idx is None or lumo_idx is None:
        print(f"Failed to identify HOMO/LUMO for {label}. Orbital occupancy
issue.")
        return False

    homo_ev = mo_energy[homo_idx] * 27.211386
    lumo_ev = mo_energy[lumo_idx] * 27.211386
    gap_ev = lumo_ev - homo_ev

    # Check for valid HOMO/LUMO energies
    if abs(gap_ev) < 1e-6:
        print(f"Warning: HOMO and LUMO energies are identical for {label}. Possible
DFT convergence issue.")
        return False

    # Compute reactivity descriptors
    hardness, softness, electronegativity, electrophilicity =
compute_reactivity_descriptors(homo_ev, lumo_ev, label)
```

```
    # Save cube files for HOMO and LUMO
    os.makedirs('/kaggle/working/orbitals', exist_ok=True)
    homo_cube_file = f'/kaggle/working/orbitals/{label}_homo.cube'
    lumo_cube_file = f'/kaggle/working/orbitals/{label}_lumo.cube'
    try:
        cubegen.orbital(pyscf_mol, homo_cube_file, mf.mo_coeff[:, homo_idx])
        cubegen.orbital(pyscf_mol, lumo_cube_file, mf.mo_coeff[:, lumo_idx])
        if not (os.path.exists(homo_cube_file) and os.path.exists(lumo_cube_file)):
            print(f"Failed to create cube files for {label}.")
            return False
    except Exception as e:
        print(f"Error generating cube files for {label}: {e}")
        return False

    # Save summary
    save_summary(label, homo_ev, lumo_ev, gap_ev, dipole, hardness, softness,
electronegativity, electrophilicity)

    # Print results
    end_time = time.time()
    print(f"{label} results (basis: {basis}):")
    print(f"HOMO: {homo_ev:.4f} eV")
    print(f"LUMO: {lumo_ev:.4f} eV")
    print(f"Gap: {gap_ev:.4f} eV")
    print(f"Time: {end_time - start_time:.2f} seconds")
    print(f"Cube files saved: {homo_cube_file}, {lumo_cube_file}\n")

    # Visualize molecule with Py3DMol (structure only)
    mol_block = Chem.MolToMolBlock(mol)
    viewer = py3Dmol.view(width=400, height=400)
    viewer.addModel(mol_block, 'mol')
    viewer.setStyle({'stick': {}})
    viewer.zoomTo()
    viewer.show()

    return True

def read_cube_file(filename):
    try:
        with open(filename, 'r') as f:
            lines = f.readlines()
        if len(lines) < 6:
            raise ValueError(f"Cube file {filename} is too short or corrupted.")
        natoms = int(lines[2].split()[0])
        origin = np.array([float(x) for x in lines[2].split()[1:4]])
        nx, ny, nz = [int(lines[i].split()[0]) for i in range(3, 6)]
        voxel = np.array([[float(lines[i].split()[j]) for j in range(1, 4)] for i
in range(3, 6)])
        if nx <= 0 or ny <= 0 or nz <= 0:
            raise ValueError(f"Invalid dimensions in cube file {filename}: nx={nx},
ny={ny}, nz={nz}")
        data = []
        for line in lines[6 + natoms:]:
            data.extend([float(x) for x in line.split()])
        data = np.array(data).reshape((nx, ny, nz))
        if np.all(data == 0):
            print(f"Warning: Volumetric data in {filename} is all zeros. Orbital
data may be invalid.")
        if np.any(np.isnan(data)) or np.any(np.isinf(data)):
            raise ValueError(f"Invalid volumetric data in {filename}: contains NaN
or Inf values.")
        x = np.linspace(0, nx-1, nx) * voxel[0][0] + origin[0]
        y = np.linspace(0, ny-1, ny) * voxel[1][1] + origin[1]
        z = np.linspace(0, nz-1, nz) * voxel[2][2] + origin[2]
        X, Y, Z = np.meshgrid(x, y, z, indexing='ij')
```

```
        return X, Y, Z, data
    except Exception as e:
        print(f"Error reading cube file {filename}: {e}")
        return None, None, None, None

def plot_2d_slice(X, Y, data, label, plane='xy', z_slice=0):
    if X is None or Y is None or data is None:
        print(f"Skipping 2D plot for {label} due to invalid cube data.")
        return
    plt.figure(figsize=(8, 6))
    try:
        if plane == 'xy':
            z_idx = np.argmin(np.abs(Z[0, 0, :] - z_slice))
            slice_data = data[:, :, z_idx]
            plt.contourf(X[:, :, z_idx], Y[:, :, z_idx], slice_data, levels=20,
cmap='RdBu')
            plt.colorbar(label='Orbital amplitude')
            plt.xlabel('X (Angstrom)')
            plt.ylabel('Y (Angstrom)')
        plt.title(f'{label} 2D Slice ({plane} plane, z={z_slice:.2f} Å)')
        os.makedirs('/kaggle/working/orbitals', exist_ok=True)
        plt.savefig(f'/kaggle/working/orbitals/{label}_2d_slice.png')
        plt.show()
    except Exception as e:
        print(f"Error plotting 2D slice for {label}: {e}")

def visualize_3d_orbital(cube_file, label, smiles, isovalues=[0.01, 0.02, 0.03]):
    try:
        with open(cube_file, 'r') as f:
            cube_data = f.read()
        viewer = py3Dmol.view(width=400, height=400)
        for isoval in isovalues:
            viewer.addVolumetricData(cube_data, 'cube', {'isoval': isoval, 'color':
'blue', 'opacity': 0.8})
            viewer.addVolumetricData(cube_data, 'cube', {'isoval': -isoval,
'color': 'red', 'opacity': 0.8})
        mol = Chem.MolFromSmiles(smiles)
        if mol is None:
            print(f"Failed to parse SMILES for {label}")
            return
        mol = Chem.AddHs(mol)
        embed_params = AllChem.ETKDGv3()
        embed_params.maxIterations = 2000
        embed_params.randomSeed = 42
        for attempt in range(5):
            embed_status = AllChem.EmbedMolecule(mol, embed_params)
            if embed_status == 0 and mol.GetConformer().Is3D():
                print(f"Embedding successful for {label} on attempt {attempt + 1}")
                break
            embed_params.randomSeed += 1
        else:
            print(f"Embedding failed for {label} after 5 attempts. Using UFF
optimization...")
            try:
                AllChem.UFFOptimizeMolecule(mol, maxIters=2000)
                if not mol.GetConformer().Is3D():
                    print(f"UFF optimization failed for {label}. Skipping 3D
structure.")
                    viewer.show()
                    return
            except Exception as e:
                print(f"UFF optimization failed for {label}: {e}. Skipping 3D
structure.")
                viewer.show()
                return
        try:
```

```
            AllChem.MMFFOptimizeMolecule(mol, maxIters=2000)
            print(f"MMFF optimization successful for {label}")
        except Exception as e:
            print(f"MMFF optimization failed for {label}: {e}. Using current
geometry.")
        mol_block = Chem.MolToMolBlock(mol)
        viewer.addModel(mol_block, 'mol')
        viewer.setStyle({'stick': {}})
        viewer.zoomTo()
        viewer.show()
        print(f"3D visualization for {label} displayed with isovalues
{isovalues}.")
    except Exception as e:
        print(f"Error in 3D visualization for {label}: {e}")

# Main execution
smiles =
r'C[C@H]1C=CC=C(C(=O)NC2=C(C(=C3C(=C2O)C(=C(C4=C3C(=O)[C@](O4)(OC=C[C@@H]([C@H]([C@
H]([C@@H]([C@@H]([C@@H]([C@H]1O)C)O)C)O)C)OC)C)C)O)O)/C=N/N5CCN(CC5)C)C'   #SMILES
of compound
label = '172967307'           #Output Folder Name

success = compute_homo_lumo_and_save(smiles, label, basis='sto-3g')

if success:
    homo_file = f'/kaggle/working/orbitals/{label}_homo.cube'
    lumo_file = f'/kaggle/working/orbitals/{label}_lumo.cube'
    mep_file = f'/kaggle/working/orbitals/{label}_mep.cube'

    # Read and plot HOMO
    X, Y, Z, homo_data = read_cube_file(homo_file)
    if homo_data is not None:
        plot_2d_slice(X, Y, homo_data, f'{label}_homo', plane='xy', z_slice=0)
        visualize_3d_orbital(homo_file, f'{label}_homo', smiles, isovalues=[0.01,
0.02, 0.03])

    # Read and plot LUMO
    X, Y, Z, lumo_data = read_cube_file(lumo_file)
    if lumo_data is not None:
        plot_2d_slice(X, Y, lumo_data, f'{label}_lumo', plane='xy', z_slice=0)
        visualize_3d_orbital(lumo_file, f'{label}_lumo', smiles, isovalues=[0.01,
0.02, 0.03])

    # Visualize MEP
    if os.path.exists(mep_file):
        visualize_3d_orbital(mep_file, f'{label}_mep', smiles, isovalues=[0.001,
0.002])

    # Debug: List files in output directory
    output_dir = '/kaggle/working/orbitals'
    print("Final files in output 'orbitals' directory:", os.listdir(output_dir) if
os.path.exists(output_dir) else "Directory not created")
else:
    print("Computation failed, skipping visualization.")
```